

FUNCTIONAL REQUIREMENTS SPECIFICATION

User Access Maintenance

BRD-11 User Access Maintenance (QPS User Access Maintenance Module)

BIBS – BDOI Broker System

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1.0	25 Sep 2026	iorta TechNXT Business Analysis	iorta TechNXT Project Manager	BDOI Product Owner (pending)	First issue for BDOI review; built access functions as the baseline, BRD changes specified from the design; aligned with the cross-BRD decisions

Distribution List

Name / Unit	Role	Organisation	Purpose
Product Owner, Marketing Business System	Approver	BDOI	Review and sign-off
Business Administrators (process owner)	Business owner	BDOI	Review of all FRs
Unit Heads, Combank and Corbank Marketing / Corporate Processing	Approver	BDOI	Review of the request and approval steps
ITIO-SRE and ITSD-AMS	System administration	BDO Unibank	Review of the System Administrator functions, sign-in and logs
ES-BPDS, Business Project Services	BRD owner	BDO Unibank ESG	Traceability check against the BRD
Project team	Delivery	iorta TechNXT	Build, test and UAT preparation

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1 Introduction

1.1 Purpose

This Functional Requirements Specification (FRS) states how BIBS (BDOI Broker System, on iNXT BrokerVerse) meets the User Access Maintenance business requirements of BDO Insurance and Reinsurance Brokers, Inc. (BDOI). It turns each BRD requirement into functional requirements with actors, flows, rules, validations, screens, fields, notifications, audit and acceptance criteria.

BDOI uses this document to confirm that user access will be managed as the business expects. The project team uses it to build, test and prepare user acceptance testing (UAT). Every functional requirement (FR) cites the BRD requirement it meets and the BRD page.

Part of user access is already built in BIBS: users, roles, permissions, sign-in with lock-out, the session policy, access requests under four eyes and the User Access Matrix (built for BRD-1 and BRD-3). This document describes those functions as the **baseline** (section 2.3) and specifies the BRD-11 changes on top of them from the build design (R3). Where a function is built, the FR quotes the screen names, messages and codes of the build. Where it is new, the message code is assigned at build and the FR gives the message text only.

1.2 Scope

The BRD was written in April 2025 for the Quotation and Pre-processing System (QPS), before BIBS. In BIBS it becomes the **system-wide** user access maintenance: every BIBS module uses the same users, group profiles (BIBS roles) and permissions (UQ19).

Table 1: Scope of this FRS

Area	In scope	Source
Sign-in and session	Log-in by persona, inactivity warning and log-out, lock-out, directory authentication (EUA / Windows ID), password policy, log-out and session log	BRD 1.001, 2.001, 3.001, 4.001; NFR p.13-14, 17
User access requests	Enrol, modify, deactivate and reactivate users; drafts, remarks, chosen approver, correction of returned requests, cancellation, view of own requests, bulk requests, effective dates	BRD 1.002 to 1.009; NFR p.13, 17
Approval	Notification, list and filter, details, approve, reject, return; second approval of risky changes	BRD 2.002; NFR p.17
Group profiles	Requests to create, modify, deactivate and reactivate group profiles with approvers in order; implementation by the System Administrator	BRD 3.002; p.6
System Administrator	Group profiles of the UAM personas; functions assigned to profiles; user data maintenance	BRD 4.002; NFR p.13-14
Reports and logs	User Access Report, User Group Profile Report, Group Profile Membership list, User Access Audit Log; structured access-change log	BRD 3.003, 4.003; p.18

Out of scope for this phase:

- Infrastructure items the BRD answers with "Follow existing QPS set up" (servers, network, DR, environments, support). They follow the BIBS deployment standards (section 8).

- Data and user migration: "N/A" in the BRD (p.12). Existing users can be loaded with the bulk request (FR-UA-019).
- Automatic joiner and leaver feeds from HR. The BRD does not ask for them; bulk requests cover mass changes.
- The final EUA, LDAP / Active Directory, single sign-on and external ACL connections. BIBS builds the seam and keeps local sign-in until BDO supplies the interfaces (UQ04, UQ14; decision D6 of R4).

1.3 References

Table 2: Reference documents

Ref.	Document	Version / date	Location
R1	Business Requirements Document - QPS User Access Maintenance Module, 39 pages (pp.1-19 text; pp.20-38 the signed scanned copy; p.39 e-mail sign-off)	v1, 15-Apr-2025; signed April-May 2025	docs/source-documents/User Access Maintenance.pdf
R2	BDOI User Access Maintenance (BRD-11) requirements baseline and fit/gap	current	docs/requirements/BDOI_UAM_BRD_SPEC.md
R3	User Access Maintenance build design	current (proposal for review)	docs/architecture/USER_ACCESS_DESIGN.md
R4	Cross-BRD decisions and answered questions	current	docs/requirements/BDOI_CROSS_BRD_DECISIONS.md
R5	Built access functions: users, roles and sign-in (security); access requests, User Access Matrix, retention (nbadmin); screens	as built	backend/.../security, backend/.../nbadmin, frontend/src/features/nbadmin, frontend/src/features/admin
R6	FRS BRD-3 Product Maintenance (role-permission change requests, PMADD05)	v1.0	docs/deliverables/out/FRS/BIBS_FRS_BRD-03_Product_Maintenance_v1.0.pdf
R7	BDO UX guidelines (brand, screen patterns)	current	docs/design/BDO_UX_GUIDELINES.md

Page references in this document ("p.8") are pages of the BRD PDF (R1). The BRD has no NFR IDs; the NFR rows carry the analyst's IDs UAM-NFR-01 to UAM-NFR-41 of R2.

1.4 Definitions and acronyms

Term	Meaning
Access request	A request to create, change, deactivate or reactivate a user, or to change a group profile, decided under four eyes; numbered AR-yyyy-nnnnnn
Approver	The user who reviews and approves, rejects or returns access requests (BRD section B)
BRD	Business Requirements Document
Business Administrator	The BDOI user who defines and manages group profiles and raises group-profile requests (BRD section C)
EUA	BDO end-user authentication service that validates the Windows ID and password (NFR p.14)
FR	Functional requirement of this document (FR-UA-nnn)
Group profile	A set of permissions assigned to a group of users (BRD p.4); a role in BIBS
LDAP / AD	Lightweight Directory Access Protocol / Active Directory

Term	Meaning
Permission	One function or capability in BIBS, with an area (module) and an action class (VIEW, CREATE, AMEND, APPROVE)
QPS	Quotation and Pre-processing System, the legacy system the BRD was written for
Requestor	The user who submits user access requests (BRD section A)
SSO	Single sign-on
System Administrator	The IT user who implements group profiles and maintains users and roles (BRD section D)
UAM-NFR-nn	ID given by the analyst to a non-functional requirement of the BRD (R2 section 7)
UAT	User acceptance testing
UQnn	Open question on BRD-11 raised by the project team (section 10.3)
User Access	The permissions and privileges granted to a user (BRD p.4)
User Access Matrix	The screen and export that show which roles hold which permissions, by permission and by action class
Windows ID	The user's BDO network log-on name, used for EUA authentication

1.5 How to read the functional requirements

Each FR in section 4 has the same parts:

- A header table with the **BRD trace** (requirement ID and page), the **actor**, the **priority** ("Must have" for the rows the BRD marks "Mandatory: Yes"; rows without a flag are noted), the **fit** class of the baseline (R2), and the **screens** and **API** that implement it. API paths marked "(designed)" are added by the build of BRD-11; the others exist today.
- **Description, preconditions, main flow and alternate and exception flows.** Where BIBS already has the function, the description starts with the baseline and then states the change.
- **Business rules.** *Configurable* rules are maintained by the System Administrator in BIBS (parameter or list of values, section 9). *Fixed* rules are part of the system and change only through a change request.
- **Validations and messages:** the check, the message the user sees and its code. Codes quoted exist in the built system. For a check added by BRD-11 the code column shows "To be assigned at build". A "-" marks a screen check (for example a blank mandatory field), which has no business code.
- **Screens and fields, notifications, audit** and numbered **acceptance criteria**, which are the basis of the BRD-11 test plan.

The BRD lists 160 requirement lines, many of them sub-steps of one function ("create request", "add remarks", "save remarks"). The FRs group them by function; section 11 lists every line with its FR.

Table 3: Fit classes (from the requirements baseline, R2)

Class	Meaning
FIT	Works today in BIBS
CONFIGURE	Needs set-up only (roles, parameters, lists)
CHANGE	Extends a built function
NEW	A function that does not exist in BIBS before BRD-11

The 160 requirement lines split into 67 FIT, 4 CONFIGURE, 61 CHANGE and 28 NEW; the 41 NFR rows into 13 FIT, 8 CONFIGURE, 12 CHANGE and 8 NEW (R2, section 3).

2 Business context and process overview

2.1 Business context

Today BDOI coordinates every QPS user access request with IT, which causes delays (p.5). The BRD asks for a module in which the business unit manages user permissions, roles and access levels itself, securely and with self-service requests, and changes roles simply. It assumes that a User Access Matrix is defined (p.5).

Table 4: Current and envisioned process (BRD p.5-6)

#	Current process (before)	Envisioned process in BIBS (after)
1	Access requests are sent to IT and coordinated by the business unit	The Requestor raises the request in BIBS, as a draft or submitted, alone or in bulk
2	The approver is reached by e-mail	The Requestor chooses the approver; BIBS notifies the approver and lists the request under "Assigned to me"
3	IT grants the access	BIBS grants the access on approval, or on the effective date
4	Group profiles are created or changed by IT on request	The Business Administrator requests the change, the approvers approve in order, and the System Administrator implements it in BIBS
5	Access lists are compiled by hand	Four reports and a structured audit log show who has which access, since when and who changed it

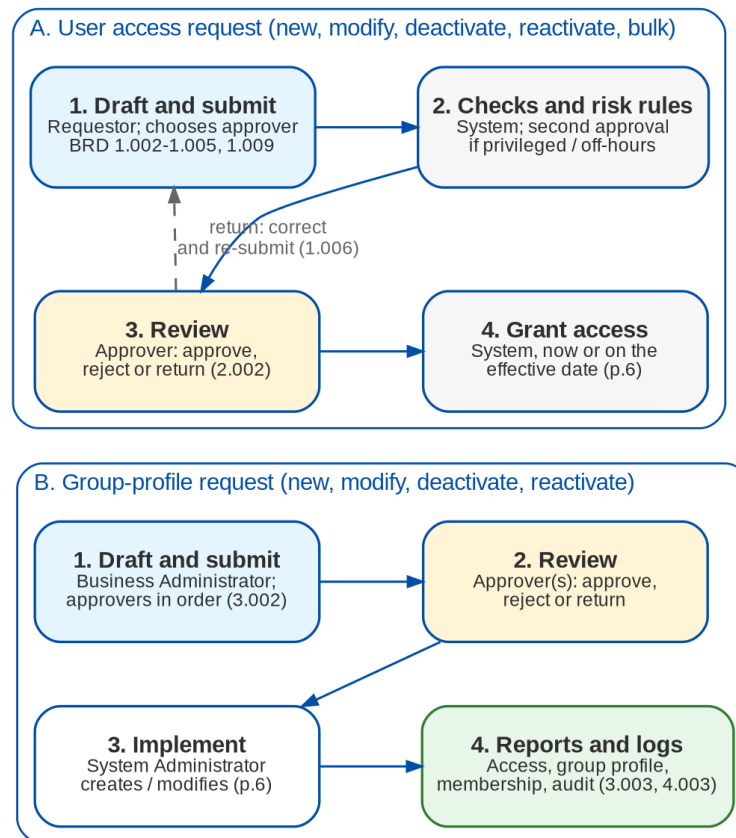
2.2 Process overview

The BRD shows two processes (p.6). The table lists the steps and Figure 1 shows them with the BIBS additions (risk rules, effective date, reports).

Table 5: Process steps

#	Step	Owner	What happens in BIBS	BRD
1	Draft	Requestor	Creates a request to enrol, modify, deactivate or reactivate a user, or a bulk file; enters data, group profiles, effective date and remarks; chooses the approver; saves, edits or cancels the draft	1.002-1.005, 1.009
2	Submit	Requestor	BIBS checks the request; status PENDING; the chosen approver is notified	1.00x.4, 2.002.1
3	Risk check	System	A change to a high-privilege profile, or a submission outside working hours, needs a second approval and raises an alert	NFR 11 (p.17)
4	Decide	Approver	Approves, rejects with a reason, or returns with remarks	2.002.5-2.002.7
5	Correct	Requestor	Corrects the returned request and re-submits it, or cancels it	1.006, 1.007
6	Apply	System	Applies the change on approval or on the effective date; logs from / to values; notifies the requester and the affected user	p.6; 1.008.1.4; NFR 10

#	Step	Owner	What happens in BIBS	BRD
7	Group profile	Business Administrator, Approvers, System Administrator	Requests a new, changed, deactivated or reactivated profile; approvers decide in order; the System Administrator implements the approved request	3.002; p.6
8	Reports and logs	Business Administrator, System Administrator, Auditor	User access, group profile, membership and audit log reports	3.003, 4.003



Actor: Requestor / Business Administrator Approver System System Administrator

Figure 1: User access request and group-profile request by actor (BRD p.6)

2.3 Baseline: access functions built today

The table lists what BIBS does today and what BRD-11 changes. The FRs refer to it as "the baseline".

Table 6: Built access functions and the BRD-11 change

Function	Built today (screen, permission)	BRD-11 change
Sign-in and lock-out	Login screen; user name and password; the account locks after LOGIN_MAX_FAILED_ATTEMPTS = 3 failed attempts until an administrator unlocks it; every success and failure is in the audit trail	Directory authentication with the Windows ID (parked seam); log-out and session log
Session policy	Inactivity warning after SESSION_IDLE_WARNING_MINUTES = 15; sign-out after SESSION_TIMEOUT_MINUTES = 30 of inactivity; warning SESSION_EXPIRY_WARNING_MINUTES = 30 before the fixed token expiry	None; confirmed by the BRD (Q42)
Users	Administration > Users (USER_MANAGE): user name, full name, e-mail, roles, authorisation limit, last login, status; unlock and reset password	Windows ID, business unit group, user level, status incl. online; direct edits limited to requests
Roles	Administration > Roles & Permissions (ROLE_MANAGE): create a role, name it, tick permissions	Active flag, description, privilege level; changes only by implementing an approved group-profile request
Access requests	Broking Setup > Access Requests (ACCESS_REQUEST to raise, ACCESS_APPROVE to decide): types Create user, Change roles, Disable user, Enable user, Change role permissions; submitted directly; status PENDING, APPROVED, REJECTED; the approver cannot be the requester; approval applies the change at once and shows a temporary password once	Drafts, return, correction, cancellation, chosen approver(s), new types (modify user, group-profile create / deactivate / reactivate), effective date, bulk, second approval, history, implementation step
Notifications	All holders of ACCESS_APPROVE on submission; the requester on the decision	The chosen approver only; returned, cancelled, second approval, implementation; the affected user
User Access Matrix	Broking Setup > User Access Matrix: roles by permission and by area / action class, Excel export	Every permission gets an area, so the group-profile report shows a module for every task
Password	My Profile > Change password: at least 10 characters with upper and lower case, digit and symbol; administrator reset	Password history, maximum and minimum age, forced change after reset, self-service reset
Audit	Insert-only audit trail; Administration > Audit Trail report (CTL-AUDIT) with summary text	Structured, insert-only access-change log with from / to values; four reports

3 Personas and roles

3.1 Personas

Table 7: *Personas and BIBS roles (design R3, section 6.2)*

Persona	BIBS role	Responsibilities in User Access Maintenance	BRD
Requestor	UAM_REQUESTOR (new)	Raises user access requests: enrol, modify, deactivate, reactivate, bulk; corrects and cancels them; views own requests	Section A (1.001-1.009)
Approver	UAM_APPROVER (new); NB_APPROVER keeps its approval right	Reviews and approves, rejects or returns requests assigned to him or her	Section B (2.001-2.002)
Second approver	UAM_SECOND_APPROVER (new)	Reviews privileged or out-of-hours changes after the first approval	NFR 11 (p.17)
Business Administrator	BUSINESS_ADMIN (exists)	Raises group-profile requests; generates the reports; supports users	Section C (3.001-3.003)
System Administrator	SYSADMIN (exists)	Implements approved group-profile requests; defines the UAM group profiles and their functions; unlocks users and resets passwords; audit logs	Section D (4.001-4.003)
Auditor	AUDITOR (exists)	Views requests and runs the reports and the audit log	NFR 6 (p.17)
System	-	Validates, grants the access, applies dated changes, logs, notifies	p.6

The BRD's stakeholders also list ITIO-SRE (System and Database Administrators), ITIO-ES (Storage Administrator) and ITSD-AMS (application support) (p.7). They support the platform and have no functional role in BIBS beyond the System Administrator. Who the 14 Requestors and 8 Approvers are is UQ01.

3.2 Permissions

BRD 4.002.2 asks that each function can be assigned to any group profile. BIBS gives each function its own permission (area USER_ACCESS). ACCESS_REQUEST stays as the umbrella permission of the roles that hold it today, so existing roles keep working.

Table 8: *User Access Maintenance permissions (area USER_ACCESS)*

Permission	Action class	Allows	BRD
UAM_ENROLL	CREATE	Enrol a new user	4.002.2.1
UAM_MODIFY	AMEND	Modify an existing user	4.002.2.2
UAM_DEACTIVATE	AMEND	Deactivate a user	4.002.2.3
UAM_REACTIVATE	AMEND	Reactivate a user	4.002.2.4
UAM_CORRECT	AMEND	Apply correction to a returned request	4.002.2.5
UAM_CANCEL	AMEND	Cancel a request	4.002.2.6
UAM_VIEW	VIEW	View requests (own; all with ACCESS_APPROVE, USER_MANAGE or AUDIT_VIEW)	4.002.2.7
ACCESS_APPROVE (exists)	APPROVE	Review and approve requests	4.002.2.8

Permission	Action class	Allows	BRD
UAM_GROUP_REQUEST	CREATE	Submit group-profile requests	4.002.2.9
UAM_REPORT_VIEW	VIEW	Generate the user access reports	4.002.2.10
UAM_SECOND_APPROVE	APPROVE	Second approval of privileged or out-of-hours changes	NFR 11
ACCESS_REQUEST (exists)	CREATE	Umbrella of the request functions for the roles that hold it today	-
USER_MANAGE, ROLE_MANAGE (exist)	AMEND	Users and Roles screens; implementation of group-profile requests	4.002.1

3.3 Permissions matrix

Table 9: Role-to-action matrix for User Access Maintenance (proposal until UQ01 / OQ48)

Permission	Requestor	Approver	Second approver	Business Admin	System Admin	NB Approver	Auditor
UAM_ENROLL	Y						
UAM_MODIFY	Y						
UAM_DEACTIVATE	Y						
UAM_REACTIVATE	Y						
UAM_CORRECT	Y			Y			
UAM_CANCEL	Y			Y			
UAM_VIEW	Y	Y	Y	Y	Y		Y
ACCESS_APPROVE		Y				Y	
UAM_SECOND_APPROVE			Y				
UAM_GROUP_REQUEST				Y			
UAM_REPORT_VIEW		Y		Y	Y		Y
ACCESS_REQUEST				Y	Y		
USER_MANAGE, ROLE_MANAGE					Y		

Segregation of duties is enforced by the system, whatever the roles grant: the requester and the user the request is about never decide it, the second approver differs from the first, and the implementer of a group-profile request is not its requester.

4 Functional requirements

4.1 Sign-in, session and password

FR-UA-001 Log in with a persona profile from a BDO-issued device

BRD trace	BRD 1.001.1 (p.8), BRD 1.001.1.1 (p.8), BRD 2.001.1 (p.8), BRD 2.001.1.1 (p.8), BRD 3.001.1 (p.9), BRD 3.001.1.1 (p.9), BRD 4.001.1 (p.9), BRD 4.001.1.1 (p.9), UAM-NFR-18 (p.14), UAM-NFR-19 (p.14)	Actor	Requestor, Approver, Business Administrator, System Administrator
Priority	Must have (4.001.x has no flag in the BRD)	Fit	FIT; CONFIGURE for the Requestor and Approver profiles (1.001.1.1, 2.001.1.1)
Screens	Login; home page with the menu of the user's roles	API	POST /api/v1/auth/login

Description

Baseline. Users sign in to BIBS in a browser with their user ID and password. BIBS checks the credentials and the account status, counts failed attempts and locks the account after LOGIN_MAX_FAILED_ATTEMPTS failed attempts (3, for every BIBS user). Every successful and failed attempt is written to the audit trail. The menu shows only the screens of the user's roles.

Change. The Requestor and Approver profiles are delivered as the roles UAM_REQUESTOR and UAM_APPROVER. Access only from BDO-issued devices is enforced by the BDO network and endpoint policy, not by BIBS.

Preconditions

- The user has an enabled BIBS account with at least one role.

Main flow

- The user opens BIBS from a BDO-issued device and enters the user ID and password.
- BIBS checks the credentials and the account status.
- BIBS opens the home page with the menu of the user's roles.

Alternate and exception flows

- Wrong credentials. BIBS refuses the log-in and counts the failed attempt.
- Third failed attempt. The account locks; the System Administrator unlocks it (Administration > Users, Unlock) or reactivates it through a request.
- Directory mode (FR-UA-003). The password is checked by EUA instead of BIBS.

Business rules

Rule	Statement	Kind	Maintained in
R1	Lock-out after LOGIN_MAX_FAILED_ATTEMPTS consecutive failures (3); applies to all users (CQ23 answered by this BRD).	Configurable	Parameter LOGIN_MAX_FAILED_ATTEMPTS
R2	A user may hold several roles; the menu is the union of their screens (NFR 1.c-d).	Fixed	-

Validations and messages

Check	Message shown	Code
User ID or password wrong	Invalid user name or password	-
Account locked	Account is locked. Contact your administrator.	-

Notifications

- None.

Audit

- Every successful and failed log-in is recorded (LOGIN, LOGIN_FAILED) with user and time.

Acceptance criteria

1. A user with the Requestor profile logs in and sees the User Access screens of the Requestor only.
2. The third wrong password locks the account and the fourth attempt is refused with "Account is locked".
3. The audit trail lists the failed and successful attempts.

FR-UA-002 Warn before the inactivity and system log-out

BRD trace	BRD 1.001.1.2 (p.8), BRD 1.001.1.3 (p.8), BRD 2.001.1.2 (p.8), BRD 2.001.1.3 (p.8), BRD 3.001.1.2 (p.9), BRD 3.001.1.3 (p.9), BRD 4.001.1.2 (p.9), BRD 4.001.1.3 (p.9), UAM-NFR-34 (p.17)	Actor	All users
Priority	Must have (4.001.x has no flag in the BRD)	Fit	FIT
Screens	Every screen (session guard dialog)	API	GET /api/v1/system/session-policy

Description

Baseline, confirmed by the BRD. After 15 minutes without activity BIBS shows a warning; after 30 minutes without activity it signs the user out. Activity in any browser tab keeps all tabs signed in. Separately, BIBS warns 30 minutes before the fixed end of the sign-in token ("For your security the system signs you out in n minute(s)...").

The BRD reads "warning prior to system triggered log out (30 minutes)" and "inactivity warning (after 15 minutes)" (p.8, p.17). The inactivity sign-out at 30 minutes with the warning at 15 minutes meets both (Q42 answered, R4).

Preconditions

- The user is signed in.

Main flow

1. The user stops working for 15 minutes; BIBS shows the inactivity warning.
2. The user continues; the warning closes and the timer restarts.

Alternate and exception flows

- No activity for 30 minutes. BIBS signs the user out and shows the log-in screen; the session ends with reason IDLE_TIMEOUT in the session log (FR-UA-004).

Business rules

Rule	Statement	Kind	Maintained in
R1	Inactivity warning after SESSION_IDLE_WARNING_MINUTES (15).	Configurable	Parameter SESSION_IDLE_WARNING_MINUTES
R2	Inactivity sign-out after SESSION_TIMEOUT_MINUTES (30).	Configurable	Parameter SESSION_TIMEOUT_MINUTES
R3	Warning SESSION_EXPIRY_WARNING_MINUTES (30) before the fixed token expiry.	Configurable	Parameter SESSION_EXPIRY_WARNING_MINUTES

Notifications

- On-screen warnings only.

Audit

- The sign-out is recorded in the session log (FR-UA-004).

Acceptance criteria

- After 15 minutes without activity the warning appears.
- After 30 minutes without activity the user is signed out.

FR-UA-003 Authenticate with the Windows ID against BDO EUA

BRD trace	UAM-NFR-11 (p.13), UAM-NFR-17 (p.14), UAM-NFR-33 (p.17)	Actor	All users; System
Priority	Must have	Fit	NEW (seam; adapter parked until UQ04)
Screens	Login	API	POST /api/v1/auth/login (unchanged contract); directory authenticator port

Description

The BRD asks that the user ID is interfaced with EUA using the Windows ID. BIBS passes the user ID and the entered password; EUA returns whether the log-on succeeded and, on failure, an error message that BIBS shows (p.14). LDAP or Active Directory authentication and single sign-on or LDAP integration are also required (p.13, p.17).

BIBS authenticates through a directory port. Parameter AUTH_MODE selects LOCAL (password held in BIBS, today) or DIRECTORY (password checked by EUA; the user is found by the Windows ID). The lock-out, audit and session behaviour are the same in both modes, and BIBS never stores the password in DIRECTORY mode.

The EUA protocol, host and messages are not in the BRD (UQ04). BIBS keeps LOCAL mode until BDO supplies them; switching is then configuration (decision D6, R4).

Preconditions

- AUTH_MODE = DIRECTORY; the user has a Windows ID in BIBS (FR-UA-052).

Main flow

- The user enters the Windows ID and password.
- BIBS passes them to EUA.
- EUA confirms the log-on; BIBS signs the user in with the BIBS roles of the user.

Alternate and exception flows

- EUA refuses. BIBS shows the EUA message, counts the failed attempt and applies the lock-out.
- The Windows ID is not linked to a BIBS user. BIBS refuses the log-in.
- EUA not reachable. BIBS refuses the log-in with a service message; a local break-glass administrator, if BDO allows one, signs in locally (UQ04).

Business rules

Rule	Statement	Kind	Maintained in
R1	Authentication mode LOCAL or DIRECTORY.	Configurable	Parameter AUTH_MODE
R2	Authorisation (roles and permissions) stays in BIBS in both modes; an external ACL is a parked seam (UQ14).	Fixed	-

Validations and messages

Check	Message shown	Code
EUA refuses the log-on	The message returned by EUA	-
Windows ID not linked to a user	Invalid user name or password	-

Notifications

- None.

Audit

- As FR-UA-001, with the authentication mode.

Acceptance criteria

1. In DIRECTORY mode, a user signs in with the Windows ID and the network password, and BIBS stores no password.
2. A wrong password shows the EUA message and counts towards the lock-out.

FR-UA-004 Log out and keep a session log

BRD trace	UAM-NFR-35 (p.17), UAM-NFR-15 (p.14)	Actor	All users; System Administrator
Priority	Must have	Fit	CHANGE
Screens	User menu (Log Out); Administration > Users (status Online)	API	POST /api/v1/auth/logout (designed)

Description

Baseline. Log-in and failed log-in are audited. Log-out is not recorded - the web client discards the token.

Change. **Log Out** ends the session in BIBS and records LOGOUT in the audit trail. Every session is kept in a session log - user, start, last activity, end and end reason (log-out, inactivity, token expiry, ended by an administrator, account locked). The user list shows a user as Online while a session is open (UQ13).

Preconditions

- The user is signed in.

Main flow

1. The user clicks **Log Out**.

2. BIBS ends the session, records LOGOUT and shows the log-in screen.

Alternate and exception flows

- A token of an ended session is used again. BIBS refuses it.

Business rules

Rule	Statement	Kind	Maintained in
R1	Last activity is updated at most every 5 minutes.	Fixed	-
R2	Single session per device is not enforced; the session log prepares it (UQ09).	Fixed	-

Notifications

- None.

Audit

- LOGOUT audited; the session log is kept with the audit trail.

Acceptance criteria

- After Log Out, the audit log report shows the log-out with user and time.
- The Users screen shows a signed-in user as Online.

FR-UA-005 Enforce the password policy and self-service reset

BRD trace	UAM-NFR-31 (p.17), UAM-NFR-36 (p.17), UAM-NFR-37 (p.17)	Actor	All users (local accounts); System Administrator
Priority	Must have	Fit	FIT (UAM-NFR-31); CHANGE (UAM-NFR-36); NEW (UAM-NFR-37)
Screens	My Profile (Change password, Details); Login (Forgot password?); Administration > Users (Reset password)	API	POST /api/v1/auth/change-password; POST /api/v1/admin/users/{id}/reset-password; password-reset flow (designed)

Description

Baseline. Users change their own password on My Profile; the password has at least 10 characters with upper and lower case letters, a digit and a symbol. The System Administrator resets a password and BIBS shows a temporary password once.

Change. BIBS also refuses the last PASSWORD_HISTORY_COUNT passwords, requires a change after PASSWORD_MAX_AGE_DAYS, refuses a second change within PASSWORD_MIN_AGE_DAYS and forces a change at the first log-in after a reset. **Forgot password?** e-mails a single-use link valid for 30 minutes. Users update their own e-mail and mobile number on My Profile (UQ17).

These rules apply to LOCAL accounts. With EUA the password belongs to BDO (UQ08).

Preconditions

- AUTH_MODE = LOCAL.

Main flow

- The user opens My Profile and enters the current and the new password twice.
- BIBS checks the policy and saves the new password.

Alternate and exception flows

- Forgot password. The user enters the user ID on the log-in screen; BIBS e-mails the reset link to the registered address; the user sets a new password.
- Password expired. At log-in BIBS asks for a new password before opening the home page.

Business rules

Rule	Statement	Kind	Maintained in
R1	Length and complexity - at least 10 characters, upper and lower case, digit, symbol.	Fixed	-
R2	History 8, maximum age 90 days, minimum age 1 day (values of the Collections NFR, to confirm, UQ08).	Configurable	Parameters PASSWORD_HISTORY_COUNT, PASSWORD_MAX_AGE_DAYS, PASSWORD_MIN_AGE_DAYS
R3	Reset links are single-use and expire after 30 minutes.	Fixed	-

Validations and messages

Check	Message shown	Code
Current password wrong	Current password is incorrect	INVALID_PASSWORD
New password too weak	must contain upper and lower case letters and a digit and a symbol	VALIDATION_FAILED
Password used before	You used this password recently. Choose another one	To be assigned at build
Changed too soon	You changed your password less than a day ago	To be assigned at build

Notifications

- PASSWORD_EXPIRY_NOTICE to users whose password expires within 7 days (job, 06:00 daily).

Audit

- Password change and reset recorded (no password value).

Acceptance criteria

- A new password equal to one of the last 8 is refused.
- After an administrator reset, the user must change the password at the next log-in.
- The reset link works once and not after 30 minutes.

4.2 User access requests (Requestor)

FR-UA-010 Draft, edit, cancel and submit an access request

BRD trace	BRD 1.002.1.1 (p.8), BRD 1.002.1.1.4-1.002.1.1.5 (p.8), BRD 1.002.1.2-1.002.1.4 (p.8), BRD 1.003.1.1 (p.8), BRD 1.003.1.1.5-1.003.1.1.6 (p.8), BRD 1.003.1.2-1.003.1.4 (p.8), BRD 1.004.1.1 (p.8), BRD 1.004.1.1.2-1.004.1.1.3 (p.8), BRD 1.004.1.2-1.004.1.4 (p.8), BRD 1.005.1 (p.8), BRD 1.005.1.1.1-1.005.1.1.2 (p.8), BRD 1.005.1.2-1.005.1.4 (p.8)	Actor	Requestor
Priority	Must have (the 'add / save remarks' lines have no flag in the BRD)	Fit	CHANGE (create, edit, cancel, save remarks); FIT (submit, add remarks)
Screens	User Access > Access Requests; New Request; Edit Request	API	POST /api/v1/nbadmin/access-requests (draft=true, designed); PUT .../access-requests/{id} (designed); POST .../access-requests/{id}/submit (designed)

Description

Baseline. The Business Administrator clicks **New Request** on Access Requests, fills the New Access Request dialog (request, user name, full name, e-mail, home branch, roles, justification) and submits it at once. The request gets a number AR-yyyy-nnnnnn and status PENDING. There is no draft.

Change. Every request type (enrol, modify, deactivate, reactivate, and the group-profile types of section 4.4) can be saved as a DRAFT. Only its creator sees and edits a draft. The creator cancels a draft (CANCELLED, kept, never deleted) or submits it. Remarks - justification, reason for deactivation or reactivation - are saved with the draft, and every remark is also kept in the request history.

On submission BIBS runs the full checks - the user exists or not, group profiles active, user ID format, approver eligible, one open request per user - evaluates the risk rules (FR-UA-034) and moves the request to PENDING.

Preconditions

- The user holds the permission of the request type (UAM_ENROLL, UAM_MODIFY, UAM_DEACTIVATE, UAM_REACTIVATE or UAM_GROUP_REQUEST) or ACCESS_REQUEST.

Main flow

- The Requestor clicks **New Request** and chooses the type.
- The Requestor enters the data (FR-UA-011 to FR-UA-014), the effective date, the approver (FR-UA-015) and the remarks.
- The Requestor clicks **Save Draft**. BIBS saves the draft with light checks (formats only).
- The Requestor opens the draft later and edits it.
- The Requestor clicks **Submit**. BIBS runs the full checks and moves the request to PENDING.

Alternate and exception flows

- Cancel draft. The Requestor clicks **Cancel Request**, enters the reason; the request becomes CANCELLED.
- A check fails on submission. BIBS keeps the draft and shows the message.

- Compatibility. The existing one-step submission (used by the Product Maintenance role-permission requests) still creates and submits in one call.

Business rules

Rule	Statement	Kind	Maintained in
R1	Request numbers are AR-<yyyy>-<nnnnnn>.	Fixed	-
R2	A draft is visible and editable only by its creator.	Fixed	-
R3	Only one open request (DRAFT excluded) per user or per group profile.	Fixed	-
R4	Remarks (justification) are mandatory on submission, up to 1000 characters.	Fixed	-
R5	Requests are never deleted; a cancelled request stays visible with its reason.	Fixed	-

Validations and messages

Check	Message shown	Code
Justification blank	Enter the justification	-
Another request for the user is pending	A request for user <user> is already waiting for approval	ACCESS_REQUEST_PENDING
Edit of a draft by another user	Only the creator can edit this draft	To be assigned at build
Cancel without reason	Enter the reason for the cancellation	-

Screens and fields: New Request (common part)

Field	Type	Mand.	Source / LOV	Validation
Request type	List	Yes	Enrol new user, Modify user, Deactivate user, Reactivate user	Fixed after first save
Effective date	Date	No	-	Today or later; blank = on approval (FR-UA-020)
Approver	List	Yes	Eligible approvers (FR-UA-015)	Required on submission
Remarks (justification)	Long text	Yes	-	Up to 1000 characters

Notifications

- On submission, the chosen approver is notified (FR-UA-070).

Audit

- Save, submit and cancel are recorded in the request history with remarks, user and time, and in the audit trail.

Acceptance criteria

- A Requestor saves an enrolment request as a draft, edits it the next day and submits it; it becomes PENDING with the same number.
- A cancelled draft stays in My Requests with status CANCELLED and its reason.
- Another Requestor cannot open the draft.

FR-UA-011 Enrol a new user

BRD trace	BRD 1.002.1 (p.8), BRD 1.002.1.1.1 (p.8), BRD 1.002.1.1.2 (p.8), UAM-NFR-13 (p.13)	Actor	Requestor
Priority	Must have	Fit	FIT (1.002.1, 1.002.1.1.2); CHANGE (1.002.1.1.1)
Screens	New Request (type Enrol new user)	API	POST /api/v1/nbadmin/access-requests (type CREATE_USER)

Description

Baseline. A Create user request carries the user name, full name, e-mail, home branch and roles. On approval BIBS creates the user and shows a temporary password once to the approver.

Change. The request also carries the Windows ID, the business unit group, the user level and an optional effective date (NFR p.13-14). The user ID follows the BDOI format (at least 10 alphanumeric characters, format a999999999, parameter USER_ID_PATTERN). Only active group profiles are offered.

Preconditions

- The user has UAM_ENROLL (or ACCESS_REQUEST).

Main flow

- The Requestor chooses **Enrol new user**.
- The Requestor enters the new user's data and selects the group profiles.
- The Requestor completes the common part (FR-UA-010) and submits.

Alternate and exception flows

- In DIRECTORY mode (FR-UA-003) no temporary password is created; the user signs in with the Windows ID.

Business rules

Rule	Statement	Kind	Maintained in
R1	User ID pattern (default <code>^[a-zA-Z][0-9]{9}\$</code> , to confirm UQ05); existing service and demo users are exempt.	Configurable	Parameter USER_ID_PATTERN
R2	Windows ID is unique across users.	Fixed	-
R3	At least one active group profile.	Fixed	-

Validations and messages

Check	Message shown	Code
User ID with wrong characters or length	The user name has 3 to 50 letters, digits, dots, dashes or underscores	ACCESS_USERNAME
User ID not in the BDOI format	The user ID must follow the format <pattern>	To be assigned at build
User already exists	User <user> already exists	ACCESS_USER_EXISTS
Full name blank	Enter the full name of the new user	ACCESS_FULL_NAME
No group profile	Select at least one role	ACCESS_ROLES
Unknown group profile	Unknown role(s): <codes>	ACCESS_UNKNOWN_ROLE
Inactive group profile	Group profile <code> is not active	To be assigned at build

Check	Message shown	Code
Windows ID already used	Windows ID <id> belongs to another user	To be assigned at build

Screens and fields: New Request (Enrol new user)

Field	Type	Mand.	Source / LOV	Validation
User ID	Text	Yes	-	USER_ID_PATTERN; unique
Windows ID	Text	Cond.	-	Unique; required in DIRECTORY mode
Full name	Text	Yes	-	Up to 120 characters
E-mail	Text	No	-	Valid e-mail; up to 120 characters
Home branch	List	No	Branches	-
Business unit group	List	No	LOV UAM_BUSINESS_UNIT	Values from BDOI (UQ05)
User level	List	No	LOV UAM_USER_LEVEL	Values from BDOI (UQ05)
Group profiles	Multi-select	Yes	Active roles	At least one

Notifications

- As FR-UA-010; on application the new user receives the access notice (FR-UA-070).

Audit

- On application, the change log records every attribute of the new user with the request number (FR-UA-064).

Acceptance criteria

- An enrolment for user ID a013000196 with the Marketing Account Officer profile is applied after approval and the user can sign in.
- An enrolment for an existing user ID is refused with ACCESS_USER_EXISTS.
- An inactive group profile is not offered.

FR-UA-012 Modify an existing user

BRD trace	BRD 1.003.1 (p.8), BRD 1.003.1.1.1 (p.8), BRD 1.003.1.1.2 (p.8), BRD 1.003.1.1.3 (p.8)	Actor	Requestor
Priority	Must have (1.003.1.1.2 has no flag in the BRD)	Fit	CHANGE (1.003.1, 1.003.1.1.2); FIT (1.003.1.1.1, 1.003.1.1.3)
Screens	New Request (type Modify user)	API	POST /api/v1/nbadmin/access-requests (type MODIFY_USER, designed); GET /api/v1/nbadmin/users

Description

Baseline. A Change roles request replaces the roles of an existing user. The user's name, e-mail and branch change only by a direct edit of the System Administrator on the Users screen.

Change. A Modify user request changes the user data (full name, e-mail, home branch, business unit group, user level, Windows ID) and / or the group profiles in one request. The Requestor searches the user by user ID, Windows ID or name; the form shows the current values next to the new ones. Change roles stays available for compatibility.

Preconditions

- The user has UAM_MODIFY (or ACCESS_REQUEST).

Main flow

- The Requestor chooses **Modify user** and searches the user.
- BIBS loads the current data and group profiles.
- The Requestor changes the data and / or the group profiles.
- The Requestor completes the common part and submits.

Business rules

Rule	Statement	Kind	Maintained in
R1	Only the changed attributes are applied; each is logged with from and to values.	Fixed	-
R2	A user never requests a change of his or her own roles.	Fixed	-

Validations and messages

Check	Message shown	Code
User not found	User <user> does not exist	ACCESS_UNKNOWN_USER
Nothing changed	The request does not change the user	To be assigned at build
Own roles	You cannot change your own roles	SELF_ROLE_CHANGE

Screens and fields: New Request (Modify user)

Field	Type	Mand.	Source / LOV	Validation
User	Look-up	Yes	Users (user ID, Windows ID, name)	Existing user
Current / new values	Pairs	No	Full name, e-mail, home branch, business unit group, user level, Windows ID	At least one change
Group profiles	Multi-select	No	Active roles	-

Notifications

- As FR-UA-010; the affected user is notified on application (FR-UA-070).

Audit

- Change log with from and to values per attribute (FR-UA-064).

Acceptance criteria

- A request that moves a user from Marketing Team Lead to Processing Team Lead is applied after approval, and the audit log report shows "Marketing Team Lead" to "Processing Team Lead".
- A request that changes the e-mail only leaves the group profiles unchanged.

FR-UA-013 Deactivate a user

BRD trace	BRD 1.004.1 (p.8), BRD 1.004.1.1.1 (p.8)	Actor	Requestor
Priority	Must have	Fit	FIT
Screens	New Request (type Deactivate user)	API	POST /api/v1/nbadmin/access-requests (type DISABLE_USER)

Description

Baseline. A Disable user request disables the account on approval; the user can no longer sign in.

Change. Only enabled users are offered. An optional reason code (list UAM_DEACTIVATION_REASON) is kept with the remarks, and an effective date allows a future deactivation, for example on the last working day.

Preconditions

- The user has UAM_DEACTIVATE (or ACCESS_REQUEST).

Main flow

1. The Requestor chooses **Deactivate user** and searches the user.
2. The Requestor selects the reason and enters the remarks.
3. The Requestor completes the common part and submits.

Business rules

Rule	Statement	Kind	Maintained in
R1	A deactivated user keeps the group profiles so a reactivation restores them.	Fixed	-
R2	Reasons - Resigned, Transferred, Long leave, Security, Others (to confirm).	Configurable	LOV UAM_DEACTIVATION_REASON

Validations and messages

Check	Message shown	Code
User not found	User <user> does not exist	ACCESS_UNKNOWN_USER
User already disabled	User <user> is already deactivated	To be assigned at build

Screens and fields: New Request (Deactivate user)

Field	Type	Mand.	Source / LOV	Validation
User	Look-up	Yes	Enabled users	-
Reason	List	No	LOV UAM_DEACTIVATION_REASON	-
Remarks	Long text	Yes	-	Up to 1000 characters

Notifications

- As FR-UA-010.

Audit

- Change log DISABLE_USER with the request number.

Acceptance criteria

1. After approval the user cannot sign in, and the User Access Report shows the deactivation with the approver.
2. A deactivation dated the 30th is applied on the 30th (FR-UA-020).

FR-UA-014 Reactivate a user

BRD trace	BRD 1.005 (p.8), BRD 1.005.1.1 (p.8)	Actor	Requestor
Priority	Must have	Fit	FIT
Screens	New Request (type Reactivate user)	API	POST /api/v1/nbadmin/access-requests (type ENABLE_USER)

Description

Baseline. An Enable user request enables a disabled account on approval.

Change. Only disabled or locked users are offered. The request can also unlock a locked account. The BRD numbers this function one level up (BRD 1.005 and 1.005.1.1 for the sub-steps); the traceability keeps the printed IDs.

Preconditions

- The user has UAM_REACTIVATE (or ACCESS_REQUEST).

Main flow

1. The Requestor chooses **Reactivate user** and searches the user.
2. The Requestor enters the reason for reactivation.
3. The Requestor completes the common part and submits.

Business rules

Rule	Statement	Kind	Maintained in
R1	The user's group profiles are restored as they were; changed profiles need a Modify user request.	Fixed	-

Validations and messages

Check	Message shown	Code
User not found	User <user> does not exist	ACCESS_UNKNOWN_USER
User already active	User <user> is already active	To be assigned at build

Screens and fields: New Request (Reactivate user)

Field	Type	Mand.	Source / LOV	Validation
User	Look-up	Yes	Disabled or locked users	-
Unlock account	Check box	No	-	Shown when the account is locked
Remarks	Long text	Yes	-	Up to 1000 characters

Notifications

- As FR-UA-010.

Audit

- Change log ENABLE_USER (and UNLOCK) with the request number.

Acceptance criteria

1. After approval the reactivated user signs in with the group profiles held before deactivation.

FR-UA-015 Choose the approver

BRD trace	BRD 1.002.1.1.3 (p.8), BRD 1.003.1.1.4 (p.8), BRD 1.004.1.1.4 (p.8), BRD 1.005.1.1.3 (p.8)	Actor	Requestor
Priority	Must have	Fit	NEW
Screens	New Request (Approver)	API	GET /api/v1/nbadmin/approvers?type=(designed)

Description

Baseline. A submitted request is notified to every holder of ACCESS_APPROVE, and any of them can decide it.

Change. The Requestor selects the approver from a drop-down of eligible approvers - enabled holders of ACCESS_APPROVE, excluding the requester and the user the request is about. The notice goes to that approver, and the request appears in that approver's My Approvals and "Assigned to me". Other approvers see it only when UAM_ANY_APPROVER is true.

Preconditions

- A request is being drafted (FR-UA-010).

Main flow

1. The Requestor opens the Approver drop-down.
2. BIBS lists the eligible approvers.
3. The Requestor selects one and submits.

Alternate and exception flows

- The chosen approver is disabled before deciding. The Requestor chooses another approver (re-submit), or any approver decides when UAM_ANY_APPROVER is true.

Business rules

Rule	Statement	Kind	Maintained in
R1	The requester and the subject user are never eligible.	Fixed	-
R2	Who may be chosen (any holder of the approval right, or the approver of the requester's unit) is UQ02.	Configurable	Parameter UAM_ANY_APPROVER

Validations and messages

Check	Message shown	Code
Approver not selected	Select the approver	-
Approver not eligible	<user> cannot approve this request	To be assigned at build

Notifications

- UAM_REQUEST_TO_APPROVE to the chosen approver.

Audit

- Chosen approver kept on the request and in the history.

Acceptance criteria

- The drop-down does not list the requester or the user the request is about.
- Only the chosen approver receives the notification and sees the request under "Assigned to me".

FR-UA-016 Apply correction on a returned request

BRD trace	BRD 1.006.1 (p.8), BRD 1.006.1.1 (p.8), BRD 1.006.1.2 (p.8), BRD 1.006.1.3 (p.8), BRD 1.006.1.4 (p.8), BRD 1.006.1.5 (p.8)	Actor	Requestor (UAM_CORRECT)
Priority	Must have	Fit	CHANGE
Screens	Access Requests (My Requests); Edit Request	API	PUT /api/v1/nbadmin/access-requests/{id}; POST .../{id}/submit (designed)

Description

Baseline. There is no return - the approver can only approve or reject.

Change. A request returned by the approver (FR-UA-033) is RETURNED. The Requestor receives a notification with the return remarks, edits the data, adds a correction remark and re-submits it to the same approver or a newly chosen one. The history keeps every round.

Preconditions

- The request is RETURNED; the user is its creator and has UAM_CORRECT (or ACCESS_REQUEST).

Main flow

- The Requestor opens the notification; the request opens.
- The Requestor reads the return remarks and edits the data.
- The Requestor adds the correction remark and clicks **Re-submit**.
- BIBS runs the checks of FR-UA-010 and moves the request to PENDING.

Alternate and exception flows

- The Requestor cancels the returned request instead (FR-UA-017).

Business rules

Rule	Statement	Kind	Maintained in
R1	A correction remark is mandatory on re-submission.	Fixed	-

Validations and messages

Check	Message shown	Code
Correction remark blank	Enter the correction remarks	-

Notifications

- UAM_REQUEST_RETURNED to the requester; UAM_REQUEST_TO_APPROVE to the approver on re-submission.

Audit

- RETURN and RESUBMIT events with remarks in the request history.

Acceptance criteria

1. A returned request shows the approver's remarks; after correction and re-submission it is PENDING again.
2. The history shows both rounds with their remarks.

FR-UA-017 Cancel a submitted request

BRD trace	BRD 1.007.1 (p.8), BRD 1.007.1.1 (p.8), BRD 1.007.1.2 (p.8), BRD 1.007.1.3 (p.8), BRD 1.007.1.4 (p.8)	Actor	Requestor (UAM_CANCEL)
Priority	Must have	Fit	CHANGE; FIT (1.007.1.1 search)
Screens	Access Requests (My Requests)	API	POST /api/v1/nbadmin/access-requests/{id}/cancel (designed)

Description

Baseline. Requests are searched by status, type and text (user, role or request number). A pending request cannot be withdrawn.

Change. The Requestor searches the request - also by requester, approver and date range - and cancels a PENDING or RETURNED request with a mandatory reason. The request becomes CANCELLED, leaves the approver's My Approvals and the approver is notified. An approved request with a future effective date (SCHEDULED) can be cancelled before its date (UQ06).

Preconditions

- The request is PENDING, PENDING_SECOND, RETURNED or SCHEDULED; the user is its creator and has UAM_CANCEL (or ACCESS_REQUEST).

Main flow

1. The Requestor searches the request and opens it.
2. The Requestor clicks **Cancel Request** and enters the reason.
3. BIBS cancels the request and notifies the approver.

Business rules

Rule	Statement	Kind	Maintained in
R1	A cancelled request is kept with its reason; it cannot be re-opened.	Fixed	-

Validations and messages

Check	Message shown	Code
Reason blank	Enter the reason for the cancellation	-
Request already decided	Request <no.> is already <status>	To be assigned at build

Notifications

- UAM_REQUEST_CANCELLED to the approver.

Audit

- CANCEL event with reason, user and time.

Acceptance criteria

1. A pending request cancelled by its creator disappears from the approver's My Approvals and shows CANCELLED with the reason.
2. An approved and applied request cannot be cancelled.

FR-UA-018 View submitted requests and their status

BRD trace	BRD 1.008.1 (p.8), BRD 1.008.1.1 (p.8), BRD 1.008.1.2 (p.8), BRD 1.008.1.3 (p.8), BRD 1.008.1.4 (p.8)	Actor	Requestor (UAM_VIEW)
Priority	Must have (1.008.1 has no flag in the BRD)	Fit	FIT
Screens	Access Requests (My Requests); request detail (History)	API	GET /api/v1/nbadmin/access-requests; GET .../access-requests/{id}; GET .../{id}/history (designed)

Description

Baseline. The Access Requests screen lists requests with number, type, change, requested by, requested on, status and decided by, with filters by status, type and text; a row opens the request detail. The requester is notified when the request is approved or rejected.

Change. A Requestor sees his or her own requests on the tab My Requests; holders of ACCESS_APPROVE, USER_MANAGE or AUDIT_VIEW see all. The detail adds a History tab with every event and remark, and the new statuses.

Preconditions

- The user has UAM_VIEW (or ACCESS_REQUEST / ACCESS_APPROVE).

Main flow

1. The Requestor opens Access Requests, tab My Requests.
2. The Requestor filters by status or type.
3. The Requestor opens a request to see its details and history.

Business rules

Rule	Statement	Kind	Maintained in
R1	Without ACCESS_APPROVE, USER_MANAGE or AUDIT_VIEW, a user sees only the requests he or she created.	Fixed	-

Screens and fields: Access Requests (columns)

Field	Type	Mand.	Source / LOV	Validation
Request	Link	-	-	AR-yyyy-nnnnnn
Type	Text	-	-	-
Change	Text	-	-	Summary of the change
Requested by / Requested	Text / Date-time	-	-	-
Approver	Text	-	-	Chosen approver
Status	Pill	-	-	DRAFT to IMPLEMENTED (section 5)
Decided by	Text	-	-	-

Notifications

- UAM_REQUEST_DECIDED to the requester on approval or rejection (FR-UA-070).

Audit

- None (read only).

Acceptance criteria

1. A Requestor sees only his or her requests, with the current status.
2. The History tab lists submission, return, re-submission and approval with remarks.

FR-UA-019 Submit a bulk request

BRD trace	BRD 1.009.1 (p.8), BRD 1.009.1.1 (p.8), BRD 1.009.1.2 (p.8), BRD 1.009.1.3 (p.8), BRD 1.009.1.4 (p.8), BRD 1.009.1.5 (p.8), BRD 1.009.1.6 (p.8), UAM-NFR-38 (p.17)	Actor	Requestor
Priority	Must have (1.009.1 has no flag in the BRD)	Fit	NEW
Screens	User Access > Bulk Request	API	POST /api/v1/nbadmin/access-requests/bulk (multipart, designed); bulk handler UAM_ACCESS_REQUEST

Description

The Requestor downloads the template, fills one row per user - action (enrol, modify, deactivate, reactivate), user ID, Windows ID, name, e-mail, branch, business unit group, user level, group profiles, effective date, remarks - and attaches it. BIBS validates every row and shows a validation report before the batch is created.

A valid file creates a draft batch with one line request per row. The Requestor adds the batch remarks, edits the draft (replaces the file or removes lines), cancels it or submits it to the chosen approver. On approval every line is applied in its own transaction; failures are listed and the other lines stand.

Preconditions

- The user has UAM_ENROLL or UAM_MODIFY (and the permissions of the actions in the file).

Main flow

1. The Requestor opens Bulk Request and downloads the template.
2. The Requestor attaches the completed file.
3. BIBS validates each row and shows the report.
4. The Requestor enters the remarks and saves the draft batch.
5. The Requestor chooses the approver and submits the batch.

Alternate and exception flows

- Rows with errors. The report lists each row and error; the Requestor corrects the file and attaches it again.
- A line fails when applied. The line is marked failed with the reason; the batch status shows the counts.

Business rules

Rule	Statement	Kind	Maintained in
R1	Each row is checked as a single request (FR-UA-011 to FR-UA-014).	Fixed	-
R2	The batch is decided as a whole; line-by-line decisions, maximum rows and mixed actions are UQ10.	Configurable	Bulk settings (UQ10)

Validations and messages

Check	Message shown	Code
Row with an unknown action	Row <n>: action <value> is not valid	To be assigned at build
Row fails a request check	Row <n>: <message of the check>	To be assigned at build
File type not allowed	The file type is not allowed	-

Screens and fields: Bulk Request

Field	Type	Mand.	Source / LOV	Validation
File	Attachment	Yes	Template UAM_ACCESS_REQUEST	XLSX or CSV
Remarks	Long text	Yes	-	Up to 1000 characters
Approver	List	Yes	Eligible approvers	-

Notifications

- As FR-UA-010; the requester receives the result of the batch application.

Audit

- Batch and each line in the request history; each applied line in the change log.

Acceptance criteria

- A file of three enrolments creates a draft batch of three lines; after approval the three users exist.
- A row with an existing user ID is reported before the batch is created.

FR-UA-020 Apply changes on an effective date

BRD trace	UAM-NFR-14 (p.13)	Actor	Requestor; System
Priority	Must have	Fit	NEW
Screens	New Request (Effective date); Access Requests (status SCHEDULED)	API	Job UAM_EFFECTIVE_CHANGES (00:05 daily)

Description

A request may carry an effective date. When it is approved before that date, the request becomes SCHEDULED and the daily job applies it on the date, after checking it again. A blank date means the change applies on approval. A scheduled request can be cancelled before its date (FR-UA-017).

Preconditions

- The request has an effective date later than the approval date.

Main flow

- The approver approves; BIBS sets the request to SCHEDULED.

- On the effective date the job re-checks and applies the change.
- BIBS sets the request to APPROVED, logs the change and notifies the requester and the affected user.

Alternate and exception flows

- The re-check fails (for example the user was deactivated meanwhile). The request stays SCHEDULED and alert UAM_SCHEDULED_APPLY_FAILED is raised to the System Administrator.

Business rules

Rule	Statement	Kind	Maintained in
R1	Effective date is today or later.	Fixed	-
R2	An end date for temporary access is not built (UQ06).	Fixed	-

Validations and messages

Check	Message shown	Code
Effective date in the past	The effective date cannot be before today	-

Notifications

- UAM_ACCESS_CHANGED to the affected user on application.

Audit

- SCHEDULE and APPLY events; change log on application.

Acceptance criteria

- A deactivation approved on the 10th with effective date the 15th is applied on the 15th.
- A failed scheduled application raises UAM_SCHEDULED_APPLY_FAILED.

4.3 Review and approval (Approver)

FR-UA-030 Receive, list and open requests for approval

BRD trace	BRD 2.002.1 (p.8), BRD 2.002.2 (p.8), BRD 2.002.2.1 (p.8), BRD 2.002.3 (p.8), BRD 2.002.4 (p.8)	Actor	Approver (ACCESS_APPROVE)
Priority	Must have	Fit	FIT (2.002.2, 2.002.3, 2.002.4); CHANGE (2.002.1, 2.002.2.1)
Screens	My Approvals; Access Requests (tab Assigned to Me); request detail	API	GET /api/v1/nbadmin/access-requests; GET .../access-requests/{id}

Description

Baseline. Pending requests appear in My Approvals for every holder of ACCESS_APPROVE (never the requester's own) and on the Access Requests screen with status PENDING. A row opens the request detail with the requested change.

Change. The approver receives a notification for each request assigned to him or her and filters the list with **Assigned to me**. The detail shows the current values next to the requested ones (user data and group profiles), the members affected (group profiles) and the history.

Preconditions

- The user has ACCESS_APPROVE.

Main flow

1. The approver opens the notification, My Approvals or the tab Assigned to Me.
2. The approver selects a request.
3. BIBS shows the details, current and requested values, risk flags and history.

Business rules

Rule	Statement	Kind	Maintained in
R1	A request appears for its chosen approver; for every approver only when UAM_ANY_APPROVER = true.	Configurable	Parameter UAM_ANY_APPROVER

Notifications

- UAM_REQUEST_TO_APPROVE on submission or re-submission.

Audit

- None (read only).

Acceptance criteria

1. The chosen approver receives a notification and finds the request under Assigned to Me.
2. The detail shows "Marketing Team Lead" as current and "Processing Team Lead" as requested for a role change.

FR-UA-031 Approve a request and grant the access

BRD trace	BRD 2.002.5 (p.8); p.6 "System to grant the access"	Actor	Approver; System
Priority	Must have	Fit	FIT
Screens	Request detail (Approve and Apply)	API	POST /api/v1/nbadmin/access-requests/{id}/approve

Description

Baseline. **Approve and Apply** applies the change at once - creates the user, replaces the roles, disables or enables the user, or changes the role permissions - records the decision and notifies the requester. For a new user BIBS shows a temporary password once; it is never stored in clear. The approver can never be the requester.

Change. The subject user can never approve either. After approval, a request with a risk flag goes to the second approver (FR-UA-034); a request with a future effective date becomes SCHEDULED (FR-UA-020); a group-profile request goes to the next approver or to the System Administrator (FR-UA-044, FR-UA-045). Every applied attribute is logged with from and to values (FR-UA-064), and the affected user is notified.

Preconditions

- The request is PENDING and assigned to the user (or UAM_ANY_APPROVER); the user has ACCESS_APPROVE and is neither the requester nor the subject user.

Main flow

1. The approver reviews the request and enters an optional comment.
2. The approver clicks **Approve and Apply**.
3. BIBS applies the change and records the decision.

4. BIBS notifies the requester and the affected user.

Business rules

Rule	Statement	Kind	Maintained in
R1	Four eyes - the requester and the subject user never decide.	Fixed	-
R2	The temporary password of a new user is shown once to the approver (LOCAL mode).	Fixed	-
R3	The applied change is checked again at approval (the user or role may have changed since submission).	Fixed	-

Validations and messages

Check	Message shown	Code
Approver is the requester	A request cannot be decided by the user who submitted it	ACCESS_FOUR_EYES
Approver is the subject user	You cannot decide a request about your own access	To be assigned at build

Notifications

- UAM_REQUEST_DECIDED to the requester; UAM_ACCESS_CHANGED to the affected user.

Audit

- AUTHORIZE in the audit trail ('Approved and applied - <change>'); APPROVE and APPLY in the history; change log rows.

Acceptance criteria

- An approved enrolment creates the user and shows the temporary password once.
- The requester cannot approve his or her own request (ACCESS_FOUR_EYES).

FR-UA-032 Reject a request

BRD trace	BRD 2.002.6 (p.8), BRD 2.002.6.1 (p.8), BRD 2.002.6.2 (p.9)	Actor	Approver
Priority	Must have (2.002.6.1-6.2 have no flag in the BRD)	Fit	FIT
Screens	Request detail (Reject)	API	POST /api/v1/nbadmin/access-requests/{id}/reject

Description

Baseline, unchanged. The approver enters the rejection reason in Comment ("Mandatory to reject; sent to the requester") and clicks **Reject**. BIBS records the decision, notifies the requester and changes nothing. The reason is also written to the request history.

Preconditions

- As FR-UA-031.

Main flow

- The approver enters the reason and clicks **Reject**.
- BIBS sets the request to REJECTED and notifies the requester.

Business rules

Rule	Statement	Kind	Maintained in
R1	A rejected request is final; the requester raises a new request.	Fixed	-

Validations and messages

Check	Message shown	Code
Reason blank	Enter the reason of the rejection	ACCESS_REJECT_REASON
Approver is the requester	A request cannot be decided by the user who submitted it	ACCESS_FOUR_EYES

Notifications

- UAM_REQUEST_DECIDED to the requester with the reason.

Audit

- REJECT in the audit trail and the history with the reason.

Acceptance criteria

1. A rejection without a reason is refused with ACCESS_REJECT_REASON.
2. The requester sees the reason on the rejected request.

FR-UA-033 Return a request to the Requestor

BRD trace	BRD 2.002.7 (p.9), BRD 2.002.7.1 (p.9)	Actor	Approver
Priority	Must have	Fit	CHANGE
Screens	Request detail (Return)	API	POST /api/v1/nbadmin/access-requests/{id}/return (designed)

Description

Baseline. There is no return.

Change. The approver clicks **Return**, enters the remarks (reason for return) and saves. The request becomes RETURNED, the remarks are saved in the history and the requester is notified to correct it (FR-UA-016). The BRD uses ID 2.002.7.1 twice, for "add remarks" and "save remarks".

Preconditions

- As FR-UA-031; also PENDING_SECOND for the second approver.

Main flow

1. The approver enters the remarks and clicks **Return**.
2. BIBS sets the request to RETURNED and notifies the requester.

Business rules

Rule	Statement	Kind	Maintained in
R1	Return remarks are mandatory.	Fixed	-

Validations and messages

Check	Message shown	Code
Remarks blank	Enter the reason for the return	-

Notifications

- UAM_REQUEST_RETURNED to the requester.

Audit

- RETURN event with remarks, user and time.

Acceptance criteria

- A returned request is RETURNED with the approver's remarks and appears in the requester's list for correction.

FR-UA-034 Escalate privileged and out-of-hours changes for a second approval

BRD trace	UAM-NFR-40 (p.17)	Actor	System; Second approver (UAM_SECOND_APPROVE)
Priority	Must have	Fit	NEW
Screens	My Approvals; Access Requests (tab Second Approval)	API	POST /api/v1/nbadmin/access-requests/{id}/second-approve (designed); alert UAM_PRIVILEGED_CHANGE

Description

The BRD asks that changes to user privileges are monitored, evaluated and escalated for additional review, for example modifications outside working hours or from low to high privilege (p.17).

Each group profile has a privilege level (LOW, STANDARD, HIGH, ADMIN). On submission and on approval, BIBS flags a request that raises a user to a HIGH or ADMIN profile (PRIVILEGE_INCREASE) or that is submitted or approved outside UAM_WORKING_HOURS (OUTSIDE_HOURS). A flagged request needs a second approval by a holder of UAM_SECOND_APPROVE other than the first approver, and raises alert UAM_PRIVILEGED_CHANGE.

Preconditions

- A request has a risk flag and the first approver approved it.

Main flow

- BIBS sets the request to PENDING_SECOND and raises the alert.
- The second approver reviews the flags and the request.
- The second approver approves; BIBS continues as FR-UA-031.

Alternate and exception flows

- The second approver rejects or returns the request (as FR-UA-032, FR-UA-033).

Business rules

Rule	Statement	Kind	Maintained in
R1	Privilege level per group profile; SYSADMIN is ADMIN; the high-privilege profiles are UQ07.	Configurable	Group profile privilege level

Rule	Statement	Kind	Maintained in
R2	Working hours 08:00-18:00 Monday to Friday (to confirm, UQ07).	Configurable	Parameter UAM_WORKING_HOURS
R3	The second approver differs from the first approver, the requester and the subject user.	Fixed	-

Validations and messages

Check	Message shown	Code
Second approver is the first approver	The second approval is given by another approver	To be assigned at build

Notifications

- UAM_SECOND_APPROVAL to the holders of UAM_SECOND_APPROVE; alert UAM_PRIVILEGED_CHANGE.

Audit

- Risk flags kept on the request; SECOND_APPROVE event.

Acceptance criteria

- A request that adds the System Administrator profile to a user needs a second approval.
- A request submitted at 21:00 on a weekday is flagged OUTSIDE_HOURS.

4.4 Group-profile requests (Business Administrator)

FR-UA-040 Request a new group profile

BRD trace	BRD 3.002.1 (p.9), BRD 3.002.1.1 (p.9), BRD 3.002.1.3 (p.9), BRD 3.002.1.4 (p.9), BRD 3.002.1.5 (p.9), BRD 3.002.1.6 (p.9), BRD 3.002.1.7 (p.9)	Actor	Business Administrator (UAM_GROUP_REQUEST)
Priority	Must have (3.002.1.6-1.7 have no flag in the BRD)	Fit	CHANGE (3.002.1, 3.002.1.1, 1.3, 1.4, 1.7); FIT (3.002.1.5, 1.6)
Screens	User Access > Group Profile Requests (New); permission picker by area and action	API	POST /api/v1/nbadmin/access-requests (type CREATE_ROLE, designed)

Description

Baseline. The System Administrator creates roles directly on Roles & Permissions. Only a change of the permissions of an existing role goes through a request (Change role permissions, BRD-3 PMADD05, R6).

Change. The Business Administrator requests a new group profile - code, name, description, privilege level and the permissions, picked by area and action class. The request follows the lifecycle of FR-UA-010 (draft, edit, cancel, submit, remarks) and the approvers of FR-UA-044. After the last approval it waits for the System Administrator (FR-UA-045).

Preconditions

- The user has UAM_GROUP_REQUEST.

Main flow

- The Business Administrator clicks **New Group Profile Request**, type Create.

2. The Business Administrator enters the profile data and picks the permissions.
3. The Business Administrator chooses the approvers, enters the remarks and submits.

Business rules

Rule	Statement	Kind	Maintained in
R1	Profile code - capital letters, digits and underscore, up to 40 characters; unique.	Fixed	-
R2	At least one permission.	Fixed	-

Validations and messages

Check	Message shown	Code
Code already used	Role <code> already exists	DUPLICATE
No permission	Select at least one permission	To be assigned at build
Unknown permission	Unknown permission(s): <codes>	ACCESS_UNKNOWN_PERMISSION
Too many permissions in one request	Split the change into several requests	ACCESS_PERMISSION_LIST_TOO_LONG

Screens and fields: Group Profile Request (Create)

Field	Type	Mand.	Source / LOV	Validation
Code	Text	Yes	-	A-Z, 0-9, _; up to 40; unique
Name	Text	Yes	-	Up to 120 characters
Description	Text	No	-	Up to 500 characters
Privilege level	List	Yes	Low, Standard, High, Admin	-
Permissions	Picker	Yes	Permissions by area and action class	At least one
Approvers	Ordered list	Yes	Eligible approvers	At least one (FR-UA-044)
Remarks	Long text	Yes	-	Up to 1000 characters

Notifications

- UAM_REQUEST_TO_APPROVE to the first approver.

Audit

- History of the request; change log CREATE_ROLE on implementation.

Acceptance criteria

1. A request for a new profile "Processing Team Lead" with its permissions reaches FOR_IMPLEMENTATION after its approvers approve.
2. A code that already exists is refused.

FR-UA-041 Request a change to a group profile

BRD trace	BRD 3.002.2 (p.9), BRD 3.002.2.1 (p.9), BRD 3.002.2.2 (p.9), BRD 3.002.2.4 (p.9), BRD 3.002.2.5 (p.9), BRD 3.002.2.6 (p.9), BRD 3.002.2.7 (p.9), BRD 3.002.2.8 (p.9)	Actor	Business Administrator
Priority	Must have (3.002.2.2, 2.7, 2.8 have no flag in the BRD)	Fit	FIT (3.002.2, 2.2, 2.6, 2.7); CHANGE (3.002.2.1, 2.4, 2.5, 2.8)
Screens	Group Profile Requests (Modify); Change role permissions fields	API	POST /api/v1/nbadmin/access-requests (type MODIFY_ROLE_PERMISSIONS)

Description

Baseline. A Change role permissions request names the role, the permissions to add and to remove and the justification; on approval the difference is applied to the role as it is at approval, so concurrent changes are not lost (R6, FR-PM-003).

Change. The Business Administrator searches the profile from the drop-down and may also change its name, description and privilege level. The request gets drafts and approvers in order, and after approval waits for the System Administrator (FR-UA-045) unless UAM_ROLE_APPLY_ON_APPROVAL is true (UQ03).

Preconditions

- The user has UAM_GROUP_REQUEST (or ACCESS_REQUEST).

Main flow

- The Business Administrator chooses Modify and selects the profile.
- The Business Administrator picks the permissions to add and to remove and edits the profile data.
- The Business Administrator chooses the approvers, enters the remarks and submits.

Business rules

Rule	Statement	Kind	Maintained in
R1	The change applied is the difference between the role at implementation and the request.	Fixed	-
R2	Implementation step after approval (p.6); switch to apply on approval.	Configurable	Parameter UAM_ROLE_APPLY_ON_APPROVAL (UQ03)

Validations and messages

Check	Message shown	Code
No role selected	Select the role to change	ACCESS_ROLE
Unknown role	Unknown role(s): <code>	ACCESS_UNKNOWN_ROLE
Permission both added and removed	Permission(s) both added and removed: <codes>	ACCESS_PERMISSION_CONFLICT
No change	The request does not change the permissions of role <code>	ACCESS_NO_PERMISSION_CHANGE
Another request for the role pending	A request for role <code> is already waiting for approval	ACCESS_REQUEST_PENDING

Screens and fields: Group Profile Request (Modify)

Field	Type	Mand.	Source / LOV	Validation
Group profile	List	Yes	Active roles	-
Permissions to add	Multi-select	No	Permissions	Not already granted
Permissions to remove	Multi-select	No	Permissions of the role	Granted to the role
Name / description / privilege level	Text / List	No	-	-
Approvers	Ordered list	Yes	Eligible approvers	-
Remarks	Long text	Yes	-	Up to 1000 characters

Notifications

- As FR-UA-040.

Audit

- Change log ROLE_PERMISSIONS with added and removed permissions on implementation.

Acceptance criteria

- A request that adds one permission and removes another is implemented with exactly that difference.
- A request that adds and removes the same permission is refused with ACCESS_PERMISSION_CONFLICT.

Built behaviour differs from the BRD. Today an approved Change role permissions request is applied by the system at approval (BRD-3 PMADD05). The BRD-11 diagram (p.6) has the System Administrator implement group-profile changes. The build adds the implementation step with UAM_ROLE_APPLY_ON_APPROVAL = false, which also changes the Product Maintenance behaviour; BDOI confirms under UQ03 (PQ17 answered, R4).

FR-UA-042 Request the deactivation of a group profile

BRD trace	BRD 3.002.3 (p.9), BRD 3.002.3.1 (p.9), BRD 3.002.3.2 (p.9), BRD 3.002.3.4 (p.9), BRD 3.002.3.5 (p.9), BRD 3.002.3.6 (p.9), BRD 3.002.3.7 (p.9), BRD 3.002.3.8 (p.9)	Actor	Business Administrator
Priority	Must have (3.002.3.2, 3.7, 3.8 have no flag in the BRD)	Fit	NEW; FIT (3.002.3.2, 3.7); CHANGE (3.002.3.8)
Screens	Group Profile Requests (Deactivate)	API	POST /api/v1/nbadmin/access-requests (type DEACTIVATE_ROLE, designed)

Description

Baseline. Roles have no active flag; a role can only be emptied of permissions.

Change. The Business Administrator requests the deactivation of an active profile. The request lists the members of the profile. When implemented, the profile is inactive - it grants nothing, is not offered on requests and keeps its permissions for a reactivation. Whether a profile with active members may be deactivated, or members must be moved first, is UQ16.

Preconditions

- The user has UAM_GROUP_REQUEST.

Main flow

1. The Business Administrator chooses Deactivate and selects an active profile.
2. BIBS shows the members.
3. The Business Administrator chooses the approvers, enters the remarks and submits.

Business rules

Rule	Statement	Kind	Maintained in
R1	An inactive profile grants no permission to its members.	Fixed	-
R2	SYSADMIN cannot be deactivated.	Fixed	-

Validations and messages

Check	Message shown	Code
Profile already inactive	Group profile <code> is already inactive	To be assigned at build
Profile has members (if UQ16 says block)	Move the <n> members of <code> before deactivating it	To be assigned at build

Notifications

- As FR-UA-040; members are notified on implementation (UAM_ACCESS_CHANGED).

Audit

- Change log DEACTIVATE_ROLE.

Acceptance criteria

1. After implementation, the members of the deactivated profile lose its screens at their next action.
2. The deactivated profile is not offered on new requests.

FR-UA-043 Request the reactivation of a group profile

BRD trace	BRD 3.002.4 (p.9), BRD 3.002.4.1 (p.9), BRD 3.002.4.2 (p.9), BRD 3.002.4.4 (p.9), BRD 3.002.4.5 (p.9), BRD 3.002.4.6 (p.9), BRD 3.002.4.7 (p.9), BRD 3.002.4.8 (p.9)	Actor	Business Administrator
Priority	Must have (3.002.4.2, 4.7, 4.8 have no flag in the BRD)	Fit	NEW; FIT (3.002.4.2, 4.7); CHANGE (3.002.4.8)
Screens	Group Profile Requests (Reactivate)	API	POST /api/v1/nbadmin/access-requests (type REACTIVATE_ROLE, designed)

Description

The Business Administrator requests the reactivation of an inactive profile, searched from the drop-down of inactive profiles. When implemented, the profile is active again with its last permissions, and its members regain them.

Preconditions

- The user has UAM_GROUP_REQUEST.

Main flow

1. The Business Administrator chooses Reactivate and selects an inactive profile.
2. The Business Administrator chooses the approvers, enters the remarks and submits.

Business rules

Rule	Statement	Kind	Maintained in
R1	The profile is restored with the permissions it had at deactivation.	Fixed	-

Validations and messages

Check	Message shown	Code
Profile already active	Group profile <code> is already active	To be assigned at build

Notifications

- As FR-UA-040.

Audit

- Change log REACTIVATE_ROLE.

Acceptance criteria

- A reactivated profile grants its permissions again to its members.

FR-UA-044 Route group-profile requests to approvers in order

BRD trace	BRD 3.002.1.2 (p.9), BRD 3.002.2.3 (p.9), BRD 3.002.3.3 (p.9), BRD 3.002.4.3 (p.9)	Actor	Business Administrator; Approvers
Priority	Not flagged in the BRD	Fit	NEW
Screens	Group Profile Requests (Approvers); My Approvals	API	Request approvers (sequence, approver, decision, remarks, time)

Description

The BRD lets the Business Administrator select "approver/s" from the drop-down. The request holds one or more approvers in order. Each approver decides in turn; the request moves to the next approver on approval. All must approve. Any approver can reject or return it. User requests have one approver (FR-UA-015).

Preconditions

- A group-profile request is being drafted.

Main flow

- The Business Administrator adds the approvers in order.
- On submission the first approver is notified.
- Each approval notifies the next approver; the last approval moves the request to FOR_IMPLEMENTATION.

Alternate and exception flows

- An approver rejects. The request is REJECTED; later approvers are not asked.
- An approver returns. The request is RETURNED; after re-submission it starts again with the first approver.

Business rules

Rule	Statement	Kind	Maintained in
R1	Approvers in sequence and all required (default until UQ02).	Fixed	-
R2	An approver appears once; the requester is never an approver.	Fixed	-

Validations and messages

Check	Message shown	Code
No approver	Add at least one approver	-
Same approver twice	<user> is already an approver of this request	To be assigned at build

Notifications

- UAM_REQUEST_TO_APPROVE to each approver in turn.

Audit

- Each approver's decision, remarks and time on the request.

Acceptance criteria

1. With two approvers, the second is notified only after the first approves.
2. A rejection by the first approver ends the request.

FR-UA-045 Implement an approved group-profile request

BRD trace	p.6 "System Administrator to create / modify group profile"; BRD 4.002.1 (p.9)	Actor	System Administrator (ROLE_MANAGE)
Priority	Must have	Fit	CHANGE
Screens	Access Requests (tab For Implementation); Administration > Roles & Permissions (Implement Request)	API	POST /api/v1/nbadmin/access-requests/{id}/implement (designed)

Description

Baseline. The System Administrator creates and edits roles freely on Roles & Permissions.

Change. Direct role maintenance is limited to implementing approved requests. The For Implementation tab lists approved group-profile requests; **Implement Request** opens the Roles screen with the approved change, and saving it applies the change and sets the request to IMPLEMENTED. Direct role edits are refused unless the audited emergency parameter UAM_DIRECT_ROLE_EDIT is true, which raises alert UAM_DIRECT_ROLE_EDIT.

Preconditions

- The request is FOR_IMPLEMENTATION; the user has ROLE_MANAGE and is not the requester.

Main flow

1. The System Administrator opens the For Implementation tab and selects the request.
2. The System Administrator clicks **Implement Request** and reviews the change.
3. The System Administrator saves; BIBS applies it, logs it and sets the request to IMPLEMENTED.

Alternate and exception flows

- Emergency. With UAM_DIRECT_ROLE_EDIT = true the System Administrator edits a role directly; the edit is audited and alerted.

Business rules

Rule	Statement	Kind	Maintained in
R1	The implementer is never the requester.	Fixed	-
R2	Emergency direct edit off by default.	Configurable	Parameter UAM_DIRECT_ROLE_EDIT

Validations and messages

Check	Message shown	Code
Direct role edit without a request	A group profile is changed only by implementing an approved request	To be assigned at build
Implementer is the requester	A request is implemented by someone other than its requester	To be assigned at build

Notifications

- UAM_FOR_IMPLEMENTATION to the System Administrators; UAM_REQUEST_DECIDED to the requester on implementation.

Audit

- IMPLEMENT event; change log of the role.

Acceptance criteria

- An approved new profile is created only when the System Administrator implements it.
- A direct role edit is refused while UAM_DIRECT_ROLE_EDIT is false.

4.5 System Administrator functions**FR-UA-050 Define the group profiles of the access personas**

BRD trace	BRD 4.002.1 (p.9), BRD 4.002.1.1 (p.9), BRD 4.002.1.2 (p.9), BRD 4.002.1.3 (p.9)	Actor	System Administrator
Priority	Must have	Fit	FIT (4.002.1, 4.002.1.3); CONFIGURE (4.002.1.1, 4.002.1.2)
Screens	Administration > Roles & Permissions	API	GET /api/v1/admin/roles

Description

The build delivers the group profiles of the BRD personas - UAM_REQUESTOR (Requestor), UAM_APPROVER (Approver) and UAM_SECOND_APPROVER - and grants the new permissions to BUSINESS_ADMIN (Business Administrator, exists) and SYSADMIN (exists), as in section 3.3. Later changes to these profiles are group-profile requests (FR-UA-040 to FR-UA-045).

Preconditions

- None (delivered with the build).

Main flow

1. The System Administrator opens Roles & Permissions.
2. BIBS lists the delivered profiles with their permissions.

Business rules

Rule	Statement	Kind	Maintained in
R1	Roles holding ACCESS_REQUEST today also receive the new request permissions, so existing users keep their functions.	Fixed	-

Notifications

- None.

Audit

- Delivered grants are in the database migration; later changes in the change log.

Acceptance criteria

1. The Requestor, Approver and Business Administrator profiles exist with the permissions of section 3.3.

FR-UA-051 Assign functions to a group profile

BRD trace	BRD 4.002.2 (p.9), BRD 4.002.2.1 to 4.002.2.7 (p.9-10), BRD 4.002.2.8 (p.10), BRD 4.002.2.9 (p.10), BRD 4.002.2.10 (p.10), UAM-NFR-32 (p.17)	Actor	System Administrator (through approved group-profile requests)
Priority	Must have	Fit	FIT (4.002.2, 4.002.2.8); CHANGE (4.002.2.1-2.7, 2.9, 2.10)
Screens	Roles & Permissions; User Access Matrix	API	Permissions of section 3.2; GET /api/v1/nbadmin/access-matrix, /by-action

Description

Baseline. Every BIBS screen, button and API call requires a permission; roles are bundles of permissions; the User Access Matrix shows them by permission and by area / action class and exports to Excel.

Change. Each access-maintenance function of BRD 4.002.2 has its own permission (section 3.2), so it can be assigned to any profile. Every permission of BIBS gets an area and action class, so the matrix and the group-profile report show a module for every task. The request endpoints accept the type-specific permission or ACCESS_REQUEST.

Preconditions

- An approved group-profile request (FR-UA-045).

Main flow

1. The System Administrator implements a request that adds, for example, UAM_CANCEL to a profile.
2. BIBS grants the function to all members of the profile.

Business rules

Rule	Statement	Kind	Maintained in
R1	Functions and permissions - section 3.2.	Fixed	-

Validations and messages

Check	Message shown	Code
Action without permission	You are not permitted to perform this action	ACCESS_DENIED

Notifications

- None.

Audit

- Change log ROLE_PERMISSIONS.

Acceptance criteria

- A profile without UAM_CANCEL does not show **Cancel Request**, and a direct call is refused with ACCESS_DENIED.
- The User Access Matrix shows UAM_ENROLL under area USER_ACCESS, action CREATE.

FR-UA-052 Maintain user data and the user ID format

BRD trace	UAM-NFR-09 (p.13), UAM-NFR-13 (p.13), UAM-NFR-15 (p.14), UAM-NFR-16 (p.14)	Actor	System Administrator; Requestor (through requests)
Priority	Must have	Fit	FIT (UAM-NFR-09); CHANGE (UAM-NFR-13, 15, 16)
Screens	Administration > Users; Lists of Values (UAM_BUSINESS_UNIT, UAM_USER_LEVEL)	API	GET /api/v1/admin/users; PUT /api/v1/admin/users/{id} (implementation and emergency only)

Description

Baseline. Users are listed with user name, full name, roles, authorisation limit, last login and status; the System Administrator unlocks accounts and resets passwords. A user can hold several roles and a role many users; accounts are disabled and re-enabled.

Change. The user record adds the Windows ID, business unit group and user level; the status shown is Active, Disabled, Locked or Online. User IDs follow USER_ID_PATTERN. Creating and editing users directly is replaced by "Raise request" for everyone except the System Administrator in an emergency (UAM_DIRECT_ROLE_EDIT). Unlock and password reset stay direct administrator actions and are logged.

Preconditions

- The user has USER_MANAGE.

Main flow

- The System Administrator opens Users and filters by status, business unit or profile.
- The System Administrator unlocks an account or resets a password, or raises a request for other changes.

Business rules

Rule	Statement	Kind	Maintained in
R1	Status is derived from the enabled flag, the lock and an open session.	Fixed	-
R2	Business unit groups and user levels are lists maintained by the Business Administrator (values UQ05).	Configurable	LOV UAM_BUSINESS_UNIT and UAM_USER_LEVEL

Validations and messages

Check	Message shown	Code
Own roles changed directly	You cannot change your own roles	SELF_ROLE_CHANGE
Unknown role	One or more roles do not exist: <codes>	UNKNOWN_ROLE

Screens and fields: Users (columns)

Field	Type	Mand.	Source / LOV	Validation
User ID / Windows ID	Text	-	-	-
Full name	Text	-	-	-
Group profiles	Text	-	-	-
Business unit group / User level	Text	-	LOVs	-
Last login	Date-time	-	-	-
Status	Pill	-	-	Active, Disabled, Locked, Online

Notifications

- None.

Audit

- Unlock and reset in the change log (UNLOCK, PASSWORD_RESET).

Acceptance criteria

1. The Users screen shows the Windows ID, business unit, user level and status of each user.
2. An unlock is listed in the audit log report with the administrator.

4.6 Reports and logs

FR-UA-060 User Access Report

BRD trace	BRD 3.003.1 (p.9), BRD 3.003.1.1 (p.9), BRD 3.003.1.1 (2nd) (p.9), BRD 3.003.1.2 to 3.003.1.6 (p.9), UAM-NFR-10 (p.13), UAM-NFR-41 (p.18)	Actor	Business Administrator; System Administrator; Auditor (UAM_REPORT_VIEW)
Priority	Must have	Fit	NEW (3.003.1.1); CHANGE (3.003.1, 1.5, 1.6); FIT (the other columns)
Screens	User Access > User Access Reports; Reports (category Control & Audit)	API	Report code UAM-USER-ACCESS

Description

The report lists users with their group profiles and who created, modified, deactivated or reactivated them, as of a date (sample A, p.18). The "created by" column shows the approver and the request number from the change log, not the session that applied the change. Layout in section 6.

Preconditions

- The user has UAM_REPORT_VIEW.

Main flow

1. The user opens User Access Reports and selects the User Access Report.

2. The user sets the as-of date and filters.
3. BIBS generates the report; the user exports it.

Business rules

Rule	Statement	Kind	Maintained in
R1	As-of date shows the access held at the end of that date, from the change log (meaning to confirm, UQ11).	Fixed	-

Validations and messages

Check	Message shown	Code
As-of date in the future	The as-of date cannot be in the future	-

Notifications

- None.

Audit

- Report runs and exports logged.

Acceptance criteria

1. The report shows user a024000000 with profile Processing Officer, created by the approver of its request and modified by the approver of the later change.
2. The report exports to PDF, XLSX and CSV.

FR-UA-061 User Group Profile Report

BRD trace	BRD 3.003.2 (p.9), BRD 3.003.2.1 to 3.003.2.6 (p.9), UAM-NFR-41 (p.18)	Actor	Business Administrator; System Administrator; Auditor
Priority	Must have	Fit	CHANGE (3.003.2, 2.2, 2.6); FIT (the other columns)
Screens	User Access Reports	API	Report code UAM-GROUP-PROFILE

Description

The report lists each group profile with the modules and tasks it can access - module = permission area, task = permission and action class - marked With Access or No Access (sample B, p.18), and the profile's created, modified, deactivated and reactivated dates and actors.

Preconditions

- The user has UAM_REPORT_VIEW.

Main flow

1. The user selects the report, the profile(s), area and active flag.
2. BIBS generates the report; the user exports it.

Business rules

Rule	Statement	Kind	Maintained in
R1	Module and task come from the permission areas and action classes; the BRD sample's QPS task names are UQ11.	Fixed	-

Notifications

- None.

Audit

- Report runs logged.

Acceptance criteria

1. For the Marketing Account Officer profile, the report lists Client Management tasks with "With Access" or "No Access".

FR-UA-062 Group Profile Membership list

BRD trace	BRD 3.003.3 (p.9), BRD 3.003.3.1 (p.9), BRD 3.003.3.1.1 (p.9), BRD 3.003.3.1.2 (p.9), UAM-NFR-10 (p.13)	Actor	Business Administrator; System Administrator; Auditor
Priority	Must have	Fit	NEW (3.003.3, 3.1); FIT (3.003.3.1.1, 3.1.2)
Screens	User Access Reports	API	Report code UAM-GROUP-MEMBERS

Description

Baseline. The User Access Matrix shows only the number of enabled users per role.

Change. The report lists the members of each profile - user name and user ID, added by and date, modified by and date - grouped by profile (sample C, p.18, titled "User Group Profile Report" in the BRD).

Preconditions

- The user has UAM_REPORT_VIEW.

Main flow

1. The user selects the report, the profile(s) and the as-of date.
2. BIBS generates the list; the user exports it.

Notifications

- None.

Audit

- Report runs logged.

Acceptance criteria

1. The list shows every member of Marketing Team Lead with the user who added each member and the date.

FR-UA-063 User Access Audit Log

BRD trace	BRD 4.003.1 (p.10), UAM-NFR-35 (p.17), UAM-NFR-41 (p.18)	Actor	System Administrator; Business Administrator; Auditor
Priority	Must have	Fit	CHANGE
Screens	User Access Reports; Administration > Audit Trail (CTL-AUDIT, exists)	API	Report code UAM-AUDIT-LOG

Description

Baseline. The Audit Trail report (CTL-AUDIT) lists audit entries by date range, user and entity type, with summary text, and exports to PDF, Excel and CSV.

Change. The User Access Audit Log lists, for a date range, each access activity - date, activity, from, to, done by, approved by, request number (sample D, p.18) - from the access-change log and the request history. Log-ins, failed log-ins and log-outs can be included. CTL-AUDIT stays.

Preconditions

- The user has UAM_REPORT_VIEW (or AUDIT_VIEW).

Main flow

- The user selects the report and the date range, user and activity.
- BIBS generates the log; the user exports it.

Business rules

Rule	Statement	Kind	Maintained in
R1	Activities - request created, approved, rejected, returned, cancelled, implemented; user created, modified, deactivated, reactivated, unlocked; password reset; group profile created, changed, deactivated, reactivated; log-in, failed log-in, log-out.	Fixed	-

Validations and messages

Check	Message shown	Code
Date to before date from	The end date must be on or after the start date	-

Notifications

- None.

Audit

- Report runs logged.

Acceptance criteria

- A profile change of User 123 appears as "Modify User Group Profile" from "Marketing Team Lead" to "Processing Team Lead" with the user who did it.
- Log-outs appear when the option is selected.

FR-UA-064 Keep a structured, insert-only access-change log

BRD trace	BRD 4.003.1 (p.10), UAM-NFR-09 (p.13), UAM-NFR-22 (p.14)	Actor	System
Priority	Must have	Fit	FIT (UAM-NFR-22); CHANGE (structured log)
Screens	Reports of section 6; request History tab	API	Access-change log (insert-only)

Description

Baseline - every change is in the insert-only audit trail as summary text. Change - every applied change also writes one row per attribute into the access-change log (time, user or profile, activity, attribute, from

value, to value, request number, done by, approved by) in the same transaction as the change. No user can change or delete it. The reports read it.

Preconditions

- None.

Main flow

1. A change is applied (approval, scheduled application, implementation, unlock, reset).
2. BIBS writes the change and the log rows in one transaction.

Business rules

Rule	Statement	Kind	Maintained in
R1	Insert-only; enforced in the database.	Fixed	-
R2	Kept at least as long as the audit trail; QPS retention values are UQ12.	Configurable	Retention rules

Notifications

- None.

Audit

- This FR is the audit.

Acceptance criteria

1. A database update of a log row is refused.
2. Each attribute of a modified user has one row with from and to values and the request number.

4.7 Notifications

FR-UA-070 Notify requesters, approvers and affected users

BRD trace	BRD 1.006.1.1 (p.8), BRD 1.008.1.4 (p.8), BRD 2.002.1 (p.8), UAM-NFR-39 (p.17)	Actor	System
Priority	Must have	Fit	CHANGE
Screens	Notifications (bell); e-mail	API	Notification events UAM_REQUEST_TO_APPROVE, UAM_REQUEST_RETURNED, UAM_REQUEST_CANCELLED, UAM_REQUEST_DECIDED, UAM_ACCESS_CHANGED, UAM_SECOND_APPROVAL, UAM_FOR_IMPLEMENTATION

Description

Baseline. On submission every holder of ACCESS_APPROVE receives "Access request <no.> to approve"; on the decision the requester receives "Access request <no.> approved / rejected", each with a link to the request.

Change. The notice goes to the chosen approver only. The requester is also notified of a return and of the application of a scheduled change; the approver of a cancellation; the second approvers and the System

Administrators of the requests waiting for them; and the affected user (in the app and by e-mail) when his or her access changes (NFR 10, p.17).

Preconditions

- None.

Main flow

1. A request event occurs.
2. BIBS notifies the users concerned in the app and, per their preferences, by e-mail.

Business rules

Rule	Statement	Kind	Maintained in
R1	Users choose in-app or e-mail per event type; the affected-user notice is always sent by e-mail too.	Configurable	Notification preferences

Notifications

- This FR is the notification set.

Audit

- Notifications logged with recipient, type and time.

Acceptance criteria

1. A returned request notifies the requester with the remarks.
2. An applied deactivation notifies the affected user by e-mail.

FR-UA-071 E-mail designated users on failed batch runs

BRD trace	UAM-NFR-24 (p.15), UAM-NFR-25 (p.15)	Actor	System; System Administrator (recipient)
Priority	Must have	Fit	CHANGE
Screens	Administration > Jobs (run history); e-mail	API	Alert JOB_FAILURE; parameter JOB_FAILURE_RECIPIENTS

Description

Baseline. Batch jobs are scheduled, run on demand from the screen, monitored and re-run; each run is logged with success or failure, and a failed run raises the in-app alert JOB_FAILURE.

Change. A failed run is also e-mailed to the designated users in parameter JOB_FAILURE_RECIPIENTS, with the job, time and error.

Preconditions

- A batch run fails.

Main flow

1. The job run fails.
2. BIBS logs the failure, raises JOB_FAILURE and e-mails the recipients.

Business rules

Rule	Statement	Kind	Maintained in
R1	Recipients.	Configurable	Parameter JOB_FAILURE_RECIPIENTS

Validations and messages

Check	Message shown	Code
Invalid e-mail in the recipients	<address> is not a valid e-mail address	-

Notifications

- E-mail to the recipients; in-app alert JOB_FAILURE.

Audit

- Job run history.

Acceptance criteria

1. A failed UAM_EFFECTIVE_CHANGES run sends an e-mail to the recipients with the error.

5 Request lifecycle

5.1 User access requests

Figure 2 shows the statuses of a user access request. The grey box (PENDING) and the end statuses APPROVED and REJECTED exist today; the others are added by BRD-11.

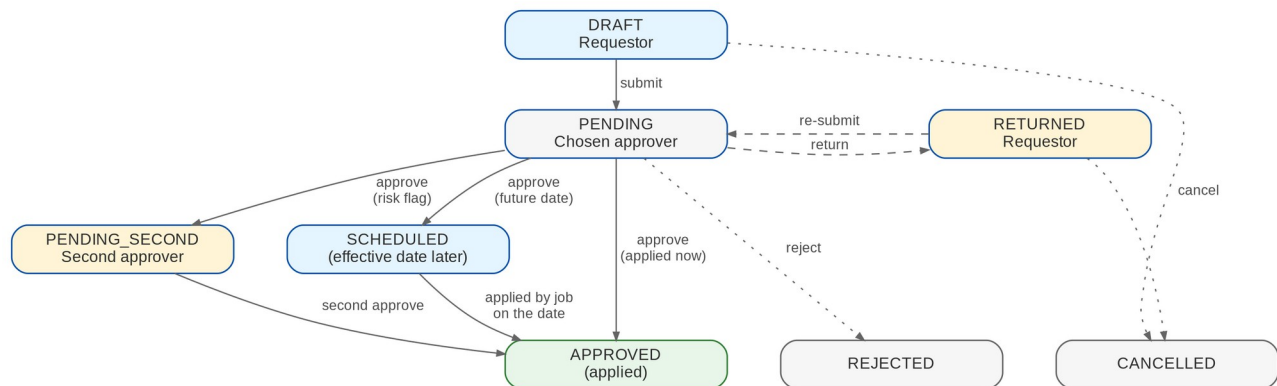


Figure 2: Status of a user access request (BRD 1.002-1.009, 2.002)

Table 10: Request statuses

Status	Meaning	Who acts
DRAFT	Saved, not submitted; visible to its creator only	Creator (edit, submit, cancel)
PENDING	Submitted; waiting for the chosen approver (group profiles - the current approver in order)	Approver; creator may cancel
PENDING_SECOND	Approved once; waiting for a second approval because of a risk flag	Second approver
RETURNED	Returned with remarks for correction	Creator (correct and re-submit, or cancel)
SCHEDULED	Approved; applied by the daily job on the effective date	System; creator or approver may cancel
APPROVED	Approved and applied	-
REJECTED	Rejected with a reason	-
CANCELLED	Cancelled by the creator with a reason	-
FOR_IMPLEMENTATION	Group profile - approved; waiting for the System Administrator	System Administrator
IMPLEMENTED	Group profile - implemented	-

Table 11: Transitions

From	Action	To	Who	Remarks
(new)	save	DRAFT	Requestor (type permission)	Optional
DRAFT	submit	PENDING	Creator	Mandatory
DRAFT	cancel	CANCELLED	Creator	Mandatory
PENDING	approve	APPROVED, SCHEDULED, PENDING_SECOND, FOR_IMPLEMENTATION or next approver	Chosen approver (ACCESS_APPROVE)	Optional

From	Action	To	Who	Remarks
PENDING_SECOND	second approve	APPROVED, SCHEDULED or FOR_IMPLEMENTATION	UAM_SECOND_APPROVE, not the first approver	Optional
PENDING, PENDING_SECOND	return	RETURNED	Approver	Mandatory
PENDING, PENDING_SECOND	reject	REJECTED	Approver	Mandatory
RETURNED	re-submit	PENDING	Creator (UAM_CORRECT)	Mandatory
PENDING, RETURNED, SCHEDULED	cancel	CANCELLED	Creator (UAM_CANCEL); approver for SCHEDULED	Mandatory
SCHEDULED	apply (job)	APPROVED	System	-
FOR_IMPLEMENTATION	implement	IMPLEMENTED	System Administrator (ROLE_MANAGE), not the requester	Optional

5.2 Group-profile requests

Figure 3 shows a group-profile request with approvers in order and the implementation step.

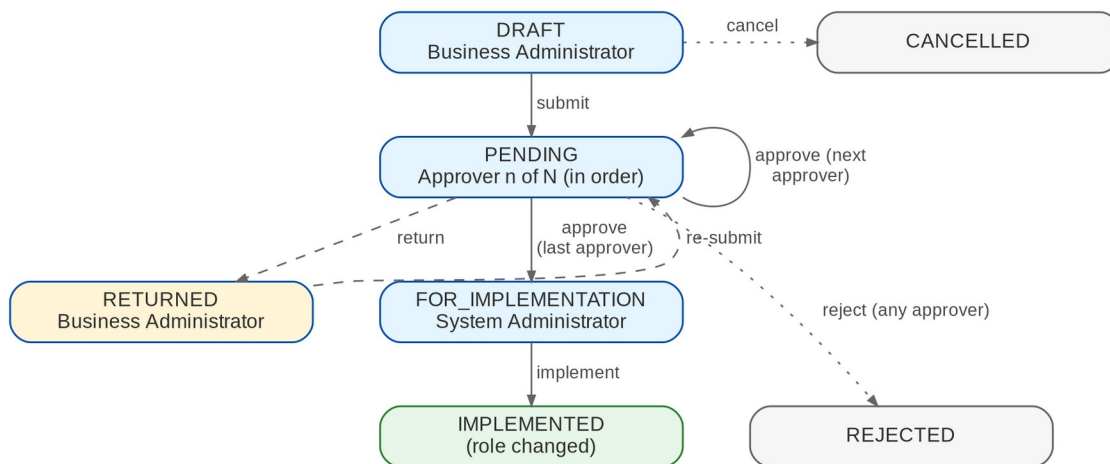


Figure 3: Status of a group-profile request (BRD 3.002; p.6)

6 Reports

6.1 Report list

Table 12: User Access Maintenance reports (category Control & Audit)

Code	Name	Purpose	BRD
UAM-USER-ACCESS	User Access Report	Users, their group profiles and who created and changed them	3.003.1; sample A
UAM-GROUP-PROFILE	User Group Profile Report	Modules and tasks each profile can access	3.003.2; sample B
UAM-GROUP-MEMBERS	Group Profile Membership	Members of each profile	3.003.3; sample C
UAM-AUDIT-LOG	User Access Audit Log	Access activities with from and to values	4.003.1; sample D
UAM-REQUESTS	Access Requests	Requests by status, type, requester, approver and age	1.008

All reports need UAM_REPORT_VIEW, export to PDF, XLSX and CSV, and print in the header the report name, the user who generated it, the date coverage ("as of" or "from / to") and the date and time generated (p.18).

6.1.1 User Access Report (UAM-USER-ACCESS)

Parameters: As of Date; Business Unit Group; Status; Group Profile. Layout: landscape, sorted by user name.

Table 13: UAM-USER-ACCESS columns

Column	Format	Content
User Name	Text	Full name
User ID / Windows ID	Text	-
User Group Profile	Text	Group profiles held on the as-of date
Business Unit / User Level	Text	-
Status	Text	Active, Disabled, Locked
Created by / Date Created	Text / Date	Approver of the enrolment (request number) and date
Modified by / Date Modified	Text / Date	Approver of the last change and date
Last Action / by	Text	Added, modified, deactivated or reactivated, and by whom

6.1.2 User Group Profile Report (UAM-GROUP-PROFILE)

Parameters: Group Profile; Area; Active. Layout: portrait, grouped by profile and module.

Table 14: UAM-GROUP-PROFILE columns

Column	Format	Content
Group Profile	Text	Profile name (and code)
Module Name	Text	Permission area
Task Name	Text	Permission and action class
Access	Text	With Access / No Access

Column	Format	Content
Date Created / Created by	Date / Text	-
Date Modified / Modified by	Date / Text	Last change and actor (added, modified, deactivated, reactivated)

6.1.3 Group Profile Membership (UAM-GROUP-MEMBERS)

Parameters: Group Profile; As of Date. Layout: portrait, grouped by profile.

Table 15: UAM-GROUP-MEMBERS columns

Column	Format	Content
Group Profile	Text	-
Member Username	Text	Full name
Member User ID	Text	-
Created by / Date Created	Text / Date	Who added the member to the profile, and when
Modified by / Date Modified	Text / Date	Last change to the membership

6.1.4 User Access Audit Log (UAM-AUDIT-LOG)

Parameters: Date From / To; User; Activity; include log-ins and log-outs (yes / no). Layout: landscape, sorted by date and time.

Table 16: UAM-AUDIT-LOG columns

Column	Format	Content
Date	Date-time	-
Activity	Text	For example "Approved Request to Enroll New User", "Modify User Group Profile of <user>"
From	Text	Value before, or "Null"
To	Text	Value after
Done By	Text	User who did it
Approved By / Request No.	Text	Where the activity came from a request

7 Interfaces and integration

Figure 4 shows the interfaces. Every BIBS module reads the effective permissions of the signed-in user; the directory, SSO and external ACL are parked seams.

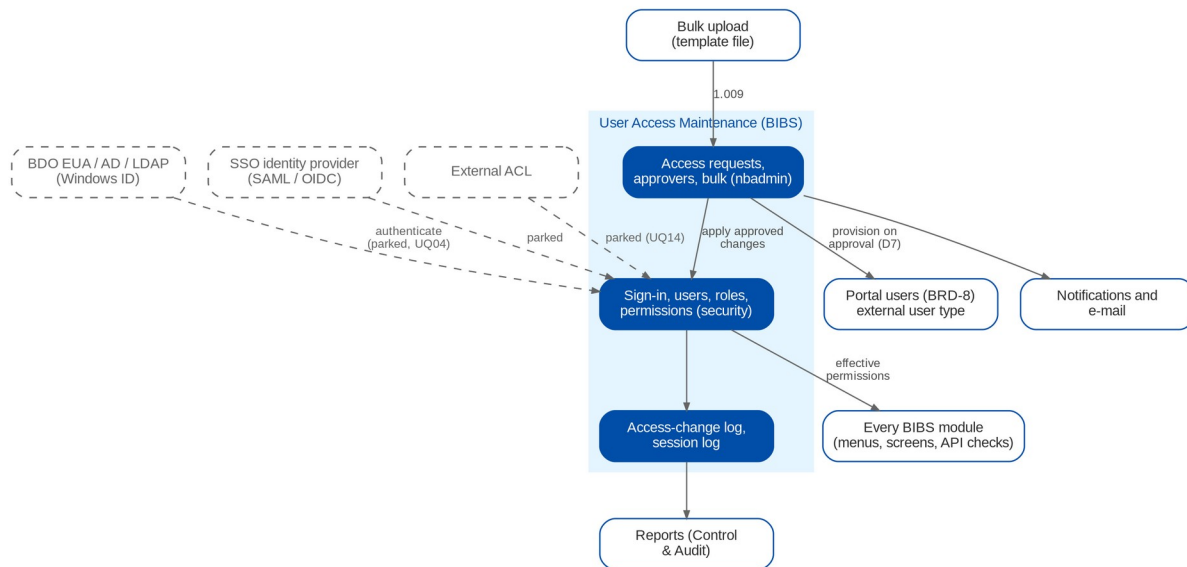


Figure 4: Interfaces of User Access Maintenance (dashed = parked)

Table 17: Interfaces

Interface	Direction	Content and trigger	BRD	Status
BDO EUA with Windows ID	Out / In	User ID and password passed at log-in; success or failure with message returned	NFR p.14	PARKED
LDAP / Active Directory	Out / In	Bind authentication, same port	NFR 1.h (p.13)	PARKED
SSO (SAML / OIDC)	In	Identity-provider assertion exchanged for a BIBS session	Other BU NFR 4 (p.17)	PARKED
External ACL	In	Authorisation by an external access-control list	NFR 1.i (p.13)	PARKED
All BIBS modules	Out	Effective permissions of the user (menus, buttons, API checks)	4.002.2	BUILT
Portal users (BRD-8 Employee Benefits)	Out	External user requests provisioned on approval (decision D7)	D7 (R4)	DESIGNED
Notifications and e-mail	Out	Request, access-change and batch-failure notices	1.006.1.1, 1.008.1.4, 2.002.1; NFR 10	BUILT
Bulk upload	In	Template file of access requests	1.009	DESIGNED
Remote log server / syslog	Out	Application and error logs	NFR p.14	PARKED

Parked seams. The EUA, LDAP / AD and SSO interfaces are required by the BRD but their protocol, host and messages are not given (UQ04). BIBS captures the Windows ID now and authenticates through a port with a LOCAL default, so connecting EUA is a configuration of a new adapter, with no change to users, roles or requests (decision D6, R4).

8 Non-functional requirements

The BRD's NFR section has no IDs; R2 numbers the rows UAM-NFR-01 to UAM-NFR-41. Most infrastructure rows are answered "Follow existing QPS set up" in the BRD; BIBS follows its own deployment standards for them. Rows that describe functions are specified as FRs in section 4 and cross-referenced here.

Table 18: Non-functional requirements (BRD p.11-18)

ID	Topic	BRD requirement	BIBS approach	FR	Fit
UAM-NFR-01	Capacity and performance	Requestor 14 users / 5 concurrent (enrol, modify, deactivate, reactivate, view: 10 tpm each; bulk 3 tpm); Approver 8 / 3 (10 tpm); BU Admin 4 / 2 (group profile 2 tpm); System Admin 1 / 1 (10 tpm, group profile 2 tpm); reports BU / System Admin 5 / 3 (10 tpm). Response 10 s, reports 20 s (p.11)	Within the BIBS sizing (145 concurrent users); online p95 under 3 seconds, reports under 20 seconds	-	FIT
UAM-NFR-02	Projected volume	10 transactions per month (average); login / logout, start-up and event reports 14 users / 5 concurrent, 10 per month, 5% growth, 10 s, 0% error; audit log report 5 / 3, 10 per month, 20% growth, 20 s, 0% error (p.11)	No specific sizing needed	-	FIT
UAM-NFR-03	Data retention (databases, logs)	Follow QPS retention policy (p.11)	Retention rules per record type; access requests, change log and session log kept at least as long as the audit trail; QPS values UQ12	-	CONFIGURE
UAM-NFR-04	Scalability, availability, reliability, DR, audit and data management, portability, interoperability, maintainability, environments, migration, support	"Follow existing QPS set up" for every item (vertical / horizontal scaling, operating hours, maintenance windows, HA, uptime, RPO / RTO, backup, DR server, delivery models, web / file / microservice integration, API gateway, ETL, core banking, monitoring, load balancer, environments Dev to DR, data and user migration, QA, training, 24/7 support) (p.11-12)	BIBS deployment standards, the same for every BRD; existing users loaded with the bulk request (FR-UA-019)	-	CONFIGURE
UAM-NFR-05	Accessibility and channels	Accessibility options for impaired vision or colour blindness; website and mobile website: "Follow existing QPS set up" (p.12)	BDO UX contrast rules; responsive screens for desktop and mobile browsers	-	FIT
UAM-NFR-06	Regulatory and compliance	MORB (BSP), Circular 808 of 2013, AMLA 2001, Data Privacy Act 2012 (p.12)	Four eyes on every access change, least privilege, insert-only audit, retention	-	FIT
UAM-NFR-07	Hardware, software, file locations, embedded IDs	Server / workstation specifications; binaries and logs in standard locations; embedded application user IDs named e_appshortname_description (e.g. p_appname_sftp) (p.12-13)	Container deployment; service accounts named by the BDO convention (for example p_bibs_db)	-	CONFIGURE
UAM-NFR-08	Network	No impact on branch / ATM operations, no change to network design; ports, devices, bandwidth, latency, interfaces (p.13)	HTTPS only; hosts and ports are configuration	-	CONFIGURE

ID	Topic	BRD requirement	BIBS approach	FR	Fit
UAM-NFR-09	User and role management 1.a-f	Online modules to view, add, modify, delete master data incl. users and roles; access via roles; more than one user per role; a user can have more than one role ; accounts can be disabled and re-enabled; maintenance recorded in the application log (p.13)	Users, roles and permissions; several roles per user and users per role; enable / disable; every change audited (CQ23 answered)	FR-UA-052, 064	FIT
UAM-NFR-10	User and role management 1.g	Query or report showing the roles assigned to users; exportable (p.13)	Reports UAM-USER-ACCESS and UAM-GROUP-MEMBERS, exportable	FR-UA-060, 062	CHANGE
UAM-NFR-11	User and role management 1.h	Supports LDAP or Active Directory authentication (p.13)	Directory authentication port; adapter parked until BDO gives the interface (Q42, UQ04)	FR-UA-003	NEW
UAM-NFR-12	User and role management 1.i	Can support authorisation of users by interfacing with an external ACL (p.13)	Seam only; parked (UQ14)	-	NEW
UAM-NFR-13	User and role management 1.j	User ID of at least ten (10) alphanumeric characters, supports the format a999999999 (p.13)	Parameter USER_ID_PATTERN checked on requests (default a999999999 format, UQ05)	FR-UA-011, 052	CHANGE
UAM-NFR-14	User and role management 1.k	Ability to specify an effective date for user and role changes (p.13)	Effective date on requests; daily job applies scheduled changes (UQ06)	FR-UA-020	NEW
UAM-NFR-15	User ID maintenance fields	Windows ID, user name, user status (active, disabled, locked out, online, etc.), Business Unit Group, User Level (p.14)	Windows ID, business unit group, user level; status Active, Disabled, Locked, Online (UQ05, UQ13)	FR-UA-052, 004	CHANGE
UAM-NFR-16	System Administrator capabilities	Access rights per group per role; user administrator maintenance; group administration; maintain reference tables; workflow set-up; screen update and maintenance (p.14)	Access rights per profile, user and group administration and reference tables exist; approver rules are parameters; screens change by release	FR-UA-052	CHANGE
UAM-NFR-17	Authentication through EUA	Interface the user ID with EUA using the Windows ID to authenticate access if the user exists in the BDO network, with password validation: the system passes the user ID and password; EUA returns whether the logon succeeded; on failure the error message is forwarded (p.14)	Windows ID sign-in through EUA when AUTH_MODE = DIRECTORY; EUA message shown; lock-out and audit unchanged (UQ04)	FR-UA-003	NEW
UAM-NFR-18	Login logging	Log all valid and invalid attempts (p.14)	Every valid and invalid log-in is audited	FR-UA-001	FIT
UAM-NFR-19	Lockout	Users are locked out after 3 invalid attempts (p.14)	LOGIN_MAX_FAILED_ATTEMPTS = 3 for every BIBS user	FR-UA-001	FIT
UAM-NFR-20	Reference / master data management	Online view / add / modify / delete; referential integrity and validation; logical deletion; changes logged; role-based access (p.14)	Lists of Values with maker-checker, effective dates and deactivation instead of delete; audited	-	FIT
UAM-NFR-21	Error logging	Error log with description, time, user ID, module ID, command / SQL; enable / disable; location; rotation; size parameters; native syslog; one line per entry; verbosity and message-type selection; transmit to a remote server (p.14)	One-line structured logs with user and module; levels changeable at run time; shipping to the BDO log server by the platform	-	CONFIGURE
UAM-NFR-22	Transaction logging and audit trail	Transaction log with type, time, user ID, module ID; enable / disable; location; transmit; all user activities logged through the audit trail and retrievable by the administrator; allow automatic save option (p.14)	Insert-only audit trail and the Audit Trail screen; meaning of 'automatic save option' is UQ15	FR-UA-064	FIT

ID	Topic	BRD requirement	BIBS approach	FR	Fit
UAM-NFR-23	Logs retention and archiving	Retention / archival / purging / backup for application, database, audit, infrastructure logs, historical data, snapshots, video, documents: "Follow existing QPS set up" (p.14)	As UAM-NFR-03 (UQ12)	-	CONFIGURE
UAM-NFR-24	Batch processing a-l	Schedule, invoke manually, monitor, re-run, graceful terminate, UI without command line, publish status, single instance, restart at the interruption point, no privileged access, multi-core, parallel independent jobs (p.15)	Job scheduler with run history, run on demand, re-run, failure alert; restart by idempotent keys; status in monitoring	FR-UA-071	CHANGE
UAM-NFR-25	Batch failure e-mail	Monitor batch runs, log success and failure; forward an e-mail notification to designated users on failed batch runs (stated twice) (p.15)	Failed runs e-mailed to JOB_FAILURE_RECIPIENTS	FR-UA-071	CHANGE
UAM-NFR-26	System monitoring; network configuration; DR	Monitor and start / stop application processes without command line; DNS aliases; configurable ports; no hosts file; standard ports; Netbackup / Bacula, clustering, offsite replication, active-active, disk estimates (p.15)	Platform monitoring, container orchestration, environment-based configuration	-	CONFIGURE
UAM-NFR-27	Data purging and archiving	Automated archiving and purging of transactional data, logs, reports, temporary files; schedule; restricted access; configurable rules (p.15-16)	Retention rules and the monthly retention review; physical archive and purge follow the BIBS retention decision (Q39, UQ12)	-	CHANGE
UAM-NFR-28	Server and workstation software	Documented install / deploy / update; deploy from repository; runs as a service; multiple instances; dynamic configuration reload; least privilege; browser only on workstations; no proprietary office software; not tied to third-party versions (p.16)	Deployed from the repository; stateless multiple instances; least-privileged database user; browser-only workstations	-	FIT
UAM-NFR-29	System documentation	Functional design, technical design, user manual, installation guide, release notes, instruction guide on how to manage users and extract the users list and group profile list; troubleshooting guide; training; capacity planning; performance benchmarks (p.16-17)	User-administration guide in the in-app Help Center and the administration guide (UQ18)	-	CHANGE
UAM-NFR-30	System development, testing and production support	Access to resources outside the BDO network, off-site / on-site development, performance testing; maintenance access (p.17)	Engagement terms	-	CONFIGURE
UAM-NFR-31	Other BU NFR 1	Password management (reset, change) (p.17)	Administrator reset and own change exist	FR-UA-005	FIT
UAM-NFR-32	Other BU NFR 2-3	Role-based access control (define roles, assign permissions, map users to roles); permission management (p.17)	Roles, permissions, user-role mapping, Roles screen and User Access Matrix exist	FR-UA-051	FIT
UAM-NFR-33	Other BU NFR 4	Integration with Single Sign-On (SSO) or LDAP (p.17)	Same directory port; SSO adapter parked (UQ04)	FR-UA-003	NEW
UAM-NFR-34	Other BU NFR 5	Session management: inactivity warning after 15 minutes; warning prior to system-triggered log out after 30 minutes (p.17)	Session policy parameters and the web session guard	FR-UA-002	FIT
UAM-NFR-35	Other BU NFR 6	Audit logging: track changes in user roles and permissions; log user login / log out; audit log reports (p.17)	Role and permission changes in the structured change log; log-out and session log added; audit log report	FR-UA-004, 063	CHANGE

ID	Topic	BRD requirement	BIBS approach	FR	Fit
UAM-NFR-36	Other BU NFR 7	Enforce strong password policies (p.17)	Complexity and length exist; history, maximum and minimum age and forced change added (UQ08)	FR-UA-005	CHANGE
UAM-NFR-37	Other BU NFR 8	Self-service and admin tools: self-service password reset; user-friendly screens to manage access; a defined user can update profile information (p.17)	Self-service reset by e-mailed single-use link (local accounts); own e-mail and mobile update (UQ08, UQ17)	FR-UA-005	NEW
UAM-NFR-38	Other BU NFR 9	Bulk uploads or batch updates of user profiles (p.17)	Bulk request (UQ10)	FR-UA-019	NEW
UAM-NFR-39	Other BU NFR 10	Notification for access changes (p.17)	Affected user notified in the app and by e-mail when the change is applied	FR-UA-070	CHANGE
UAM-NFR-40	Other BU NFR 11	Configurable workflow for access requests and approvals; changes to user privileges systematically monitored, evaluated and, if necessary, escalated for additional review (e.g. modifications outside working hours, low to high privilege) (p.17)	Privilege level per profile; privilege increase or out-of-hours change needs a second approval and raises an alert (UQ07)	FR-UA-034	NEW
UAM-NFR-41	Report requirements (section 7)	Samples A User Access Report, B User Group Profile Report, C Group Profile membership (titled "User Group Profile Report"), D User Access Audit Log (date, activity, from, to, done by); generated by, date coverage, date and time generated (p.18)	Four reports with generated by, coverage and time generated in the header	FR-UA-060 to 063	CHANGE

9 Configuration items

The items below are changed in BIBS without a release. Changes to parameters and lists are audited.

9.1 Parameters and jobs

Table 19: *User Access Maintenance parameters and jobs*

Parameter / job	Default	Meaning
LOGIN_MAX_FAILED_ATTEMPTS (exists)	3	Failed attempts before lock-out; all users
SESSION_IDLE_WARNING_MINUTES (exists)	15	Inactivity warning
SESSION_TIMEOUT_MINUTES (exists)	30	Inactivity sign-out
SESSION_EXPIRY_WARNING_MINUTES (exists)	30	Warning before the fixed token expiry
AUTH_MODE	LOCAL	LOCAL or DIRECTORY (EUA)
USER_ID_PATTERN	^[a-zA-Z][0-9]{9}\$	User ID format (to confirm, UQ05)
PASSWORD_HISTORY_COUNT	8	Previous passwords refused (UQ08)
PASSWORD_MAX_AGE_DAYS	90	Password expiry (UQ08)
PASSWORD_MIN_AGE_DAYS	1	Minimum days between changes (UQ08)
UAM_WORKING_HOURS	08:00-18:00, MON-FRI	Working hours for the out-of-hours flag (UQ07)
UAM_ANY_APPROVER	false	Any approver may decide, not only the chosen one (UQ02)
UAM_DIRECT_ROLE_EDIT	false	Emergency direct role edit (audited, alerted)
UAM_ROLE_APPLY_ON_APPROVAL	false	Group-profile requests applied on approval instead of implemented (UQ03)
JOB_FAILURE_RECIPIENTS	empty	E-mail recipients of failed batch runs
Job UAM_EFFECTIVE_CHANGES	00:05 daily	Applies scheduled requests
Job PASSWORD_EXPIRY_NOTICE	06:00 daily	Notifies users whose password expires within 7 days (LOCAL mode)

9.2 Lists of values

Table 20: *Lists of values*

List	Values delivered
UAM_BUSINESS_UNIT	Empty until BDOI supplies the business unit groups (UQ05)
UAM_USER_LEVEL	Empty until BDOI supplies the user levels (UQ05)
UAM_DEACTIVATION_REASON	Resigned; Transferred; Long leave; Security; Others (to confirm)
Group profile privilege level	Low; Standard; High; Admin

10 Assumptions, dependencies and open questions

10.1 Assumptions

Table 21: Assumptions

ID	Assumption	Related
A-UA-01	The QPS BRD applies to BIBS as a whole: every module and persona	UQ19
A-UA-02	Group profiles are BIBS roles; a user may hold several (NFR 1.c-d)	CQ23 (answered)
A-UA-03	BDOI has a User Access Matrix (p.5); its content per unit is supplied separately	OQ48
A-UA-04	Group-profile requests are implemented by the System Administrator after approval, as on p.6	UQ03
A-UA-05	Approvers of a group-profile request decide in order and all must approve	UQ02
A-UA-06	Local sign-in continues until BDO supplies the EUA interface	UQ04, D6
A-UA-07	Infrastructure answers "Follow existing QPS set up" are met by the BIBS deployment standards	Section 8
A-UA-08	Duplicate IDs 2.002.7.1 and 3.003.1.1 are kept as printed and marked "(2nd)"	R2 section 10

10.2 Dependencies

Table 22: Dependencies

ID	Dependency	Needed for
D-UA-01	BDO supplies the EUA / LDAP / SSO interface, hosts and certificates	FR-UA-003 (UQ04)
D-UA-02	BDOI supplies the business unit groups, user levels and the user ID format	FR-UA-011, 052 (UQ05)
D-UA-03	BDOI names the Requestors, Approvers and high-privilege profiles	Section 3, FR-UA-034 (UQ01, UQ07)
D-UA-04	BDOI supplies the bulk file layout and limits	FR-UA-019 (UQ10)
D-UA-05	The e-mail relay of the BIBS environment is available	FR-UA-005, 070, 071
D-UA-06	The Employee Benefits portal implements the provisioning port for external users	Section 7 (D7)

10.3 Open questions

Table 23: Open questions on BRD-11 (R2 section 9; status from R4)

ID	Question	Affects	Status
UQ01	Who are the 14 Requestors and 8 Approvers; is the Requestor separate from the Business Administrator	Section 3	OPEN
UQ02	Which users may be chosen as approver; several approvers in sequence or in parallel, all required	FR-UA-015, 044	OPEN
UQ03	Must the System Administrator implement group profiles by hand, or may BIBS apply them on approval	FR-UA-041, 045	OPEN
UQ04	EUA interface: protocol, host, Windows ID format, messages; break-glass administrator; password policy with EUA	FR-UA-003	OPEN
UQ05	Values of business unit group and user level; user ID format and relation to the Windows ID	FR-UA-011, 052	OPEN
UQ06	Effective date: start only or also end (temporary access); cancelling a scheduled request	FR-UA-020	OPEN

ID	Question	Affects	Status
UQ07	Which profiles are high privilege; working hours; who does the additional review	FR-UA-034	OPEN
UQ08	Password values for local accounts; channel of the self-service reset	FR-UA-005	OPEN
UQ09	Single session per device; single-role rule dropped	FR-UA-004	PARTIAL
UQ10	Bulk file layout, maximum rows, batch or line decisions, mixed actions	FR-UA-019	OPEN
UQ11	Module / task of sample B; meaning of "as of"	FR-UA-060, 061	OPEN
UQ12	QPS retention values for requests, audit and log-in logs	FR-UA-064	OPEN
UQ13	Real-time online status; administrator ending a session	FR-UA-004, 052	OPEN
UQ14	Which external ACL system is meant	FR-UA-003	OPEN
UQ15	Meaning of "allow automatic save option" in the audit-trail requirement	Section 8	OPEN
UQ16	Deactivation of a profile with active members	FR-UA-042	OPEN
UQ17	Profile fields users update themselves; approval of the change	FR-UA-005	OPEN
UQ18	The truncated documentation requirement on p.17	Section 8	OPEN
UQ19	The QPS BRD applies to BIBS as a whole	Section 1.2	OPEN

BRD-11 answers questions raised on other BRDs: Q42 (directory sign-in required; inactivity sign-out at 30 minutes with a 15-minute warning), CQ23 (lock-out at 3 for all users; several roles per user) and PQ17 (group-profile changes are approved and then implemented by the System Administrator) (R4, decisions D5 and D6).

11 Traceability

Every BRD-11 requirement line is met by at least one FR. The BRD prints 160 lines; IDs 2.002.7.1 and 3.003.1.1 appear twice and the second occurrence is marked "(2nd)". "Flag" repeats the BRD's Pilot Phase / Mandatory columns ("-" = not flagged). The NFR rows are traced in section 8.

Table 24: BRD requirement line to FR (160 lines)

BRD ID	Page	Activity and requirement	Flag	FR	Fit
BRD 1.001.1	p.8	Log in as a Requestor - Access the application using any BDO-issued device	Yes	FR-UA-001	FIT
BRD 1.001.1.1	p.8	Log in as a Requestor - Log in with a Requestor user profile	Yes	FR-UA-001	CONFIGURE
BRD 1.001.1.2	p.8	Log in as a Requestor - Receive the inactivity warning (15 minutes)	Yes	FR-UA-002	FIT
BRD 1.001.1.3	p.8	Log in as a Requestor - Receive a warning prior to the system-triggered log out (30 minutes)	Yes	FR-UA-002	FIT
BRD 1.002.1	p.8	Enroll New User - Submit a new User Access request	Yes	FR-UA-011	FIT
BRD 1.002.1.1	p.8	Enroll New User - 1. Create request	Yes	FR-UA-010, FR-UA-011	CHANGE
BRD 1.002.1.1.1	p.8	Enroll New User - 1.1 Input new user data	Yes	FR-UA-011	CHANGE
BRD 1.002.1.1.2	p.8	Enroll New User - 1.2 Select User Access Group profile	Yes	FR-UA-011	FIT
BRD 1.002.1.1.3	p.8	Enroll New User - 1.3 Select approver from the drop-down	Yes	FR-UA-015	NEW
BRD 1.002.1.1.4	p.8	Enroll New User - 1.4 Add remarks (i.e. justification, etc.)	-	FR-UA-010	FIT
BRD 1.002.1.1.5	p.8	Enroll New User - 1.5 Save remarks	-	FR-UA-010	CHANGE
BRD 1.002.1.2	p.8	Enroll New User - 2. Edit drafted request	Yes	FR-UA-010	CHANGE
BRD 1.002.1.3	p.8	Enroll New User - 3. Cancel drafted request	Yes	FR-UA-010	CHANGE
BRD 1.002.1.4	p.8	Enroll New User - 4. Submit request	Yes	FR-UA-010	FIT
BRD 1.003.1	p.8	Modify Existing User - Submit a modify User Access / permission request	Yes	FR-UA-012	CHANGE
BRD 1.003.1.1	p.8	Modify Existing User - 1. Create request	Yes	FR-UA-010, FR-UA-012	CHANGE
BRD 1.003.1.1.1	p.8	Modify Existing User - 1.1 Search existing user	Yes	FR-UA-012	FIT
BRD 1.003.1.1.2	p.8	Modify Existing User - 1.2 Edit / update selected user data	-	FR-UA-012	CHANGE
BRD 1.003.1.1.3	p.8	Modify Existing User - 1.3 Select User Access Group profile	Yes	FR-UA-012	FIT
BRD 1.003.1.1.4	p.8	Modify Existing User - 1.4 Select approver from the drop-down	Yes	FR-UA-015	NEW
BRD 1.003.1.1.5	p.8	Modify Existing User - 1.5 Add remarks (i.e. justification, etc.)	-	FR-UA-010	FIT
BRD 1.003.1.1.6	p.8	Modify Existing User - 1.6 Save remarks	-	FR-UA-010	CHANGE
BRD 1.003.1.2	p.8	Modify Existing User - 2. Edit drafted request	Yes	FR-UA-010	CHANGE
BRD 1.003.1.3	p.8	Modify Existing User - 3. Cancel drafted request	Yes	FR-UA-010	CHANGE
BRD 1.003.1.4	p.8	Modify Existing User - 4. Submit request	Yes	FR-UA-010	FIT
BRD 1.004.1	p.8	Deactivate User - Submit a User Access deactivation request	Yes	FR-UA-013	FIT
BRD 1.004.1.1	p.8	Deactivate User - 1. Create a request	Yes	FR-UA-010, FR-UA-013	CHANGE
BRD 1.004.1.1.1	p.8	Deactivate User - 1.1 Search an existing user	Yes	FR-UA-013	FIT
BRD 1.004.1.1.2	p.8	Deactivate User - 1.2 Add remarks (i.e. reason for deactivation)	-	FR-UA-010, FR-UA-013	FIT
BRD 1.004.1.1.3	p.8	Deactivate User - 1.3 Save remarks	Yes	FR-UA-010, FR-UA-013	CHANGE

BRD ID	Page	Activity and requirement	Flag	FR	Fit
BRD 1.004.1.1.4	p.8	Deactivate User - 1.4 Select approver from the drop-down	Yes	FR-UA-015	NEW
BRD 1.004.1.2	p.8	Deactivate User - 2. Edit drafted request	Yes	FR-UA-010	CHANGE
BRD 1.004.1.3	p.8	Deactivate User - 3. Cancel drafted request	Yes	FR-UA-010	CHANGE
BRD 1.004.1.4	p.8	Deactivate User - 4. Submit request	Yes	FR-UA-010	FIT
BRD 1.005	p.8	Reactivate User - Submit a User Access reactivation request	Yes	FR-UA-014	FIT
BRD 1.005.1	p.8	Reactivate User - 1. Create a request	Yes	FR-UA-010, FR-UA-014	CHANGE
BRD 1.005.1.1	p.8	Reactivate User - 1.1 Search an existing user	Yes	FR-UA-014	FIT
BRD 1.005.1.1.1	p.8	Reactivate User - 1.2 Add remarks (i.e. reason for reactivation)	-	FR-UA-010, FR-UA-014	FIT
BRD 1.005.1.1.2	p.8	Reactivate User - 1.3 Save remarks	Yes	FR-UA-010, FR-UA-014	CHANGE
BRD 1.005.1.1.3	p.8	Reactivate User - 1.4 Select approver from the drop-down	Yes	FR-UA-015	NEW
BRD 1.005.1.2	p.8	Reactivate User - 2. Edit a drafted request	Yes	FR-UA-010	CHANGE
BRD 1.005.1.3	p.8	Reactivate User - 3. Cancel a drafted request	Yes	FR-UA-010	CHANGE
BRD 1.005.1.4	p.8	Reactivate User - 4. Submit a request	Yes	FR-UA-010	FIT
BRD 1.006.1	p.8	Apply Correction - Apply correction on returned requests	Yes	FR-UA-016	CHANGE
BRD 1.006.1.1	p.8	Apply Correction - 1. Receive notification on the returned request	Yes	FR-UA-016, FR-UA-070	CHANGE
BRD 1.006.1.2	p.8	Apply Correction - 2. Edit data on the returned request	Yes	FR-UA-016	CHANGE
BRD 1.006.1.3	p.8	Apply Correction - 3. Add remarks	Yes	FR-UA-016	CHANGE
BRD 1.006.1.4	p.8	Apply Correction - 4. Save remarks	Yes	FR-UA-016	CHANGE
BRD 1.006.1.5	p.8	Apply Correction - 5. Re-submit request	Yes	FR-UA-016	CHANGE
BRD 1.007.1	p.8	Cancel a Request - Cancel a submitted request for approval	Yes	FR-UA-017	CHANGE
BRD 1.007.1.1	p.8	Cancel a Request - 1. Search request	Yes	FR-UA-017	FIT
BRD 1.007.1.2	p.8	Cancel a Request - 2. Add remarks (i.e. reason for cancellation, etc.)	Yes	FR-UA-017	CHANGE
BRD 1.007.1.3	p.8	Cancel a Request - 3. Save remarks	Yes	FR-UA-017	CHANGE
BRD 1.007.1.4	p.8	Cancel a Request - 4. Cancel submitted request	Yes	FR-UA-017	CHANGE
BRD 1.008.1	p.8	View Request/s - View the submitted request/s	-	FR-UA-018	FIT
BRD 1.008.1.1	p.8	View Request/s - 1. View the list of the submitted request/s	Yes	FR-UA-018	FIT
BRD 1.008.1.2	p.8	View Request/s - 2. View the status of the submitted request/s	Yes	FR-UA-018	FIT
BRD 1.008.1.3	p.8	View Request/s - 3. View the details of the submitted request/s	Yes	FR-UA-018	FIT
BRD 1.008.1.4	p.8	View Request/s - 4. Receive notification on the approved / rejected request	Yes	FR-UA-018, FR-UA-070	FIT
BRD 1.009.1	p.8	Bulk Request - Submit request for bulk creation / modification / deactivation / reactivation	-	FR-UA-019	NEW
BRD 1.009.1.1	p.8	Bulk Request - 1. Attach a file for bulk processing	Yes	FR-UA-019	NEW
BRD 1.009.1.2	p.8	Bulk Request - 2. Add remarks	Yes	FR-UA-019	NEW
BRD 1.009.1.3	p.8	Bulk Request - 3. Save remarks	Yes	FR-UA-019	NEW
BRD 1.009.1.4	p.8	Bulk Request - 4. Edit a drafted request	Yes	FR-UA-019	NEW
BRD 1.009.1.5	p.8	Bulk Request - 5. Cancel a drafted request	Yes	FR-UA-019	NEW
BRD 1.009.1.6	p.8	Bulk Request - 6. Submit a request	Yes	FR-UA-019	NEW
BRD 2.001.1	p.8	Log in as an Approver - Access the application using any BDO-issued device	Yes	FR-UA-001	FIT

BRD ID	Page	Activity and requirement	Flag	FR	Fit
BRD 2.001.1.1	p.8	Log in as an Approver - Log in with an Approver user profile	Yes	FR-UA-001	CONFIGURE
BRD 2.001.1.2	p.8	Log in as an Approver - Receive the inactivity warning (15 minutes)	Yes	FR-UA-002	FIT
BRD 2.001.1.3	p.8	Log in as an Approver - Receive a warning prior to the system-triggered log out (30 minutes)	Yes	FR-UA-002	FIT
BRD 2.002.1	p.8	Review and Approve Request/s - a. Receive notification on the request for approval	Yes	FR-UA-030, FR-UA-015, FR-UA-070	CHANGE
BRD 2.002.2	p.8	Review and Approve Request/s - b. View the list of requests for approval	Yes	FR-UA-030	FIT
BRD 2.002.2.1	p.8	Review and Approve Request/s - 1. Filter the list assigned to me	Yes	FR-UA-030	CHANGE
BRD 2.002.3	p.8	Review and Approve Request/s - c. Select request	Yes	FR-UA-030	FIT
BRD 2.002.4	p.8	Review and Approve Request/s - d. View request details	Yes	FR-UA-030	FIT
BRD 2.002.5	p.8	Review and Approve Request/s - e. Approve request	Yes	FR-UA-031	FIT
BRD 2.002.6	p.8	Review and Approve Request/s - f. Reject request	Yes	FR-UA-032	FIT
BRD 2.002.6.1	p.8	Review and Approve Request/s - 1. Add remarks (i.e. rejection reason, etc.)	-	FR-UA-032	FIT
BRD 2.002.6.2	p.9	Review and Approve Request/s - 2. Save remarks	-	FR-UA-032	FIT
BRD 2.002.7	p.9	Review and Approve Request/s - g. Return request to Requestor	Yes	FR-UA-033	CHANGE
BRD 2.002.7.1	p.9	Review and Approve Request/s - 1. Add remarks (i.e. reason for return, etc.)	Yes	FR-UA-033	CHANGE
BRD 2.002.7.1 (2nd)	p.9	Review and Approve Request/s - 2. Save remarks	Yes	FR-UA-033	CHANGE
BRD 3.001.1	p.9	Log in as a Business Administrator - Access the application using any BDO-issued device	Yes	FR-UA-001	FIT
BRD 3.001.1.1	p.9	Log in as a Business Administrator - Log in with a Business Administrator user profile	Yes	FR-UA-001	FIT
BRD 3.001.1.2	p.9	Log in as a Business Administrator - Receive the inactivity warning (15 minutes)	Yes	FR-UA-002	FIT
BRD 3.001.1.3	p.9	Log in as a Business Administrator - Receive a warning prior to the system-triggered log out (30 minutes)	Yes	FR-UA-002	FIT
BRD 3.002.1	p.9	Submit a User Access Group Profile Request - a. Submit a new User Access Group Profiles request	Yes	FR-UA-040	CHANGE
BRD 3.002.1.1	p.9	Submit a User Access Group Profile Request - 1. Create a request	Yes	FR-UA-040	CHANGE
BRD 3.002.1.2	p.9	Submit a User Access Group Profile Request - 2. Select approver/s from the drop-down	-	FR-UA-044	NEW
BRD 3.002.1.3	p.9	Submit a User Access Group Profile Request - 3. Edit a drafted request	Yes	FR-UA-040	CHANGE
BRD 3.002.1.4	p.9	Submit a User Access Group Profile Request - 4. Cancel a drafted request	Yes	FR-UA-040	CHANGE
BRD 3.002.1.5	p.9	Submit a User Access Group Profile Request - 5. Submit a request	Yes	FR-UA-040	FIT
BRD 3.002.1.6	p.9	Submit a User Access Group Profile Request - 6. Add remarks	-	FR-UA-040	FIT
BRD 3.002.1.7	p.9	Submit a User Access Group Profile Request - 7. Save remarks	-	FR-UA-040	CHANGE
BRD 3.002.2	p.9	Submit a User Access Group Profile Request - b. Submit a modify User Access Group Profiles request	Yes	FR-UA-041	FIT
BRD 3.002.2.1	p.9	Submit a User Access Group Profile Request - 1. Create a request	Yes	FR-UA-041	CHANGE

BRD ID	Page	Activity and requirement	Flag	FR	Fit
BRD 3.002.2.2	p.9	Submit a User Access Group Profile Request - 2. Search existing group profile from the drop-down	-	FR-UA-041	FIT
BRD 3.002.2.3	p.9	Submit a User Access Group Profile Request - 3. Select approver/s from the drop-down	-	FR-UA-044	NEW
BRD 3.002.2.4	p.9	Submit a User Access Group Profile Request - 4. Edit a drafted request	Yes	FR-UA-041	CHANGE
BRD 3.002.2.5	p.9	Submit a User Access Group Profile Request - 5. Cancel a drafted request	Yes	FR-UA-041	CHANGE
BRD 3.002.2.6	p.9	Submit a User Access Group Profile Request - 6. Submit a request	Yes	FR-UA-041	FIT
BRD 3.002.2.7	p.9	Submit a User Access Group Profile Request - 7. Add remarks	-	FR-UA-041	FIT
BRD 3.002.2.8	p.9	Submit a User Access Group Profile Request - 8. Save remarks	-	FR-UA-041	CHANGE
BRD 3.002.3	p.9	Submit a User Access Group Profile Request - c. Submit a User Access Group Profiles deactivation request	Yes	FR-UA-042	NEW
BRD 3.002.3.1	p.9	Submit a User Access Group Profile Request - 1. Create a request	Yes	FR-UA-042	NEW
BRD 3.002.3.2	p.9	Submit a User Access Group Profile Request - 2. Search existing group profile from the drop-down	-	FR-UA-042	FIT
BRD 3.002.3.3	p.9	Submit a User Access Group Profile Request - 3. Select approver/s from the drop-down	-	FR-UA-044	NEW
BRD 3.002.3.4	p.9	Submit a User Access Group Profile Request - 4. Edit a drafted request	Yes	FR-UA-042	NEW
BRD 3.002.3.5	p.9	Submit a User Access Group Profile Request - 5. Cancel a drafted request	Yes	FR-UA-042	NEW
BRD 3.002.3.6	p.9	Submit a User Access Group Profile Request - 6. Submit a request	Yes	FR-UA-042	NEW
BRD 3.002.3.7	p.9	Submit a User Access Group Profile Request - 7. Add remarks	-	FR-UA-042	FIT
BRD 3.002.3.8	p.9	Submit a User Access Group Profile Request - 8. Save remarks	-	FR-UA-042	CHANGE
BRD 3.002.4	p.9	Submit a User Access Group Profile Request - d. Submit a User Access Group Profiles reactivation request	Yes	FR-UA-043	NEW
BRD 3.002.4.1	p.9	Submit a User Access Group Profile Request - 1. Create a request	Yes	FR-UA-043	NEW
BRD 3.002.4.2	p.9	Submit a User Access Group Profile Request - 2. Search existing group profile from the drop-down	-	FR-UA-043	FIT
BRD 3.002.4.3	p.9	Submit a User Access Group Profile Request - 3. Select approver/s from the drop-down	-	FR-UA-044	NEW
BRD 3.002.4.4	p.9	Submit a User Access Group Profile Request - 4. Edit a drafted request	Yes	FR-UA-043	NEW
BRD 3.002.4.5	p.9	Submit a User Access Group Profile Request - 5. Cancel a drafted request	Yes	FR-UA-043	NEW
BRD 3.002.4.6	p.9	Submit a User Access Group Profile Request - 6. Submit a request	Yes	FR-UA-043	NEW
BRD 3.002.4.7	p.9	Submit a User Access Group Profile Request - 7. Add remarks	-	FR-UA-043	FIT
BRD 3.002.4.8	p.9	Submit a User Access Group Profile Request - 8. Save remarks	-	FR-UA-043	CHANGE
BRD 3.003.1	p.9	Generate Reports - a. Generate customized report	Yes	FR-UA-060	CHANGE
BRD 3.003.1.1	p.9	Generate Reports - 1. User Access Report	Yes	FR-UA-060	NEW
BRD 3.003.1.1 (2nd)	p.9	Generate Reports - 1.1 User name	Yes	FR-UA-060	FIT

BRD ID	Page	Activity and requirement	Flag	FR	Fit
BRD 3.003.1.2	p.9	Generate Reports - 1.2 Group profile	Yes	FR-UA-060	FIT
BRD 3.003.1.3	p.9	Generate Reports - 1.3 Date created	Yes	FR-UA-060	FIT
BRD 3.003.1.4	p.9	Generate Reports - 1.4 Date modified	Yes	FR-UA-060	FIT
BRD 3.003.1.5	p.9	Generate Reports - 1.5 Created by	Yes	FR-UA-060	CHANGE
BRD 3.003.1.6	p.9	Generate Reports - 1.6 Added / Modified / deactivated / reactivated by	Yes	FR-UA-060	CHANGE
BRD 3.003.2	p.9	Generate Reports - 2. User Group Profile Report	Yes	FR-UA-061	CHANGE
BRD 3.003.2.1	p.9	Generate Reports - 2.1 Group profile name	Yes	FR-UA-061	FIT
BRD 3.003.2.2	p.9	Generate Reports - 2.2 Modules accessed under each group profile	Yes	FR-UA-061	CHANGE
BRD 3.003.2.3	p.9	Generate Reports - 2.3 Date created	Yes	FR-UA-061	FIT
BRD 3.003.2.4	p.9	Generate Reports - 2.4 Date modified	Yes	FR-UA-061	FIT
BRD 3.003.2.5	p.9	Generate Reports - 2.5 Created by	Yes	FR-UA-061	FIT
BRD 3.003.2.6	p.9	Generate Reports - 2.6 Added / Modified / deactivated / reactivated by	Yes	FR-UA-061	CHANGE
BRD 3.003.3	p.9	Generate Reports - 3. Group Profile Membership list	Yes	FR-UA-062	NEW
BRD 3.003.3.1	p.9	Generate Reports - 1. List of members under each profile	Yes	FR-UA-062	NEW
BRD 3.003.3.1.1	p.9	Generate Reports - 1.1 Group profile	Yes	FR-UA-062	FIT
BRD 3.003.3.1.2	p.9	Generate Reports - 1.2 User name	Yes	FR-UA-062	FIT
BRD 4.001.1	p.9	Log in as a System Administrator - Access the application using any BDO-issued device	-	FR-UA-001	FIT
BRD 4.001.1.1	p.9	Log in as a System Administrator - Log in with a System Administrator user profile	-	FR-UA-001	FIT
BRD 4.001.1.2	p.9	Log in as a System Administrator - Receive the inactivity warning (15 minutes)	-	FR-UA-002	FIT
BRD 4.001.1.3	p.9	Log in as a System Administrator - Receive a warning prior to the system-triggered log out (30 minutes)	-	FR-UA-002	FIT
BRD 4.002.1	p.9	Manage Users of the System - a. Define user profile / role	Yes	FR-UA-050, FR-UA-045	FIT
BRD 4.002.1.1	p.9	Manage Users of the System - 1. Create user profile / role for Requestor	Yes	FR-UA-050	CONFIGURE
BRD 4.002.1.2	p.9	Manage Users of the System - 2. Create user profile / role for Approver	Yes	FR-UA-050	CONFIGURE
BRD 4.002.1.3	p.9	Manage Users of the System - 3. Create user profile / role for Business Administrator	Yes	FR-UA-050	FIT
BRD 4.002.2	p.9	Manage Users of the System - b. Define specific functionality or capability for a specific user profile / role	Yes	FR-UA-051	FIT
BRD 4.002.2.1	p.9	Manage Users of the System - 1. Assign Enroll New User to a specific user profile / role	Yes	FR-UA-051	CHANGE
BRD 4.002.2.2	p.9	Manage Users of the System - 2. Assign Modify Existing User function to a specific user profile / role	Yes	FR-UA-051	CHANGE
BRD 4.002.2.3	p.9	Manage Users of the System - 3. Assign Deactivate User function to a specific user profile / role	Yes	FR-UA-051	CHANGE
BRD 4.002.2.4	p.9	Manage Users of the System - 4. Assign Reactivate User function to a specific user profile / role	Yes	FR-UA-051	CHANGE
BRD 4.002.2.5	p.9	Manage Users of the System - 5. Assign Apply Correction function to a specific user profile / role	Yes	FR-UA-051	CHANGE
BRD 4.002.2.6	p.9	Manage Users of the System - 6. Assign Cancel a Request function to a specific user profile / role	Yes	FR-UA-051	CHANGE
BRD 4.002.2.7	p.10	Manage Users of the System - 7. Assign View Request/s function to a specific user profile / role	Yes	FR-UA-051	CHANGE

BRD ID	Page	Activity and requirement	Flag	FR	Fit
BRD 4.002.2.8	p.10	Manage Users of the System - 8. Assign Review and Approve Request/s function to a specific user profile / role	Yes	FR-UA-051	FIT
BRD 4.002.2.9	p.10	Manage Users of the System - 9. Assign Submit a User Access Group Profiles Request function to a specific user profile / role	Yes	FR-UA-051	CHANGE
BRD 4.002.2.10	p.10	Manage Users of the System - 10. [Assign] Generate Reports Request function to a specific user profile / role	Yes	FR-UA-051	CHANGE
BRD 4.003.1	p.10	Generate Logs - a. Generate audit logs for compliance and audit purposes as needed	Yes	FR-UA-063, FR-UA-064	CHANGE

12 Sign-off

By signing, BDOI confirms that this FRS describes the User Access Maintenance functions it expects in BIBS, and accepts the assumptions in section 10.1. Open questions in section 10.3 stay open; their answers are applied as configuration or through a change request.

Name	Role	Organisation	Signature	Date
	Product Owner, Marketing Business System	BDOI		
	Process Owner, User Access Maintenance	BDOI		
	Unit Head, Combank / Corbank Marketing and Corporate Processing	BDOI		
	Head, Comptrollership	BDOI		
	Program Manager, ES-BPS	BDO Unibank ESG		
	Project Manager	iorta TechNXT		